

Does Entry Remedy Collusion? Evidence from the Generic Prescription Drug Cartel[†]

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Entry represents a fundamental threat to cartels. We study the extent and effect of this behavior in the largest price-fixing case in US history, which involves generic drug manufacturing. We link information on the cartel's internal operations to regulatory filings and market data. There is a substantial increase in entry after cartel formation, but regulatory approvals delay most entrants by two to four years. We estimate a structural model and simulate counterfactual equilibria. Absent entry, cartel profits would be dramatically higher. Correspondingly, reducing regulatory delays by just 1–2 years equates to consumer compensating variation of \$612 million–\$1.5 billion. (JEL L12, L13, L25, L51, L65, M31, O34)

Cartels thwart competition, even in modern economies. The US Department of Justice has prosecuted price-fixing by all 3 major canned tuna brands, 70 auto parts suppliers, 15 foreign exchange trading institutions, over 100 real estate investors bidding in foreclosure auctions, and many others in the last few years alone. However, as cartels raise prices and profits, they may also attract uncooperative entrants, whose efforts to gain market share will undercut the incumbents' agreements. Hence, entry can serve as a fundamental safeguard against sustained collusion. In many markets, though, entry is a slow and expensive process, so the likelihood that entry restores competitive prices depends critically on barriers to entry and the length of delays. We study the extent and effect of entry on cartelized markets in the context of the largest price-fixing case in US history, which involves generic prescription drug manufacturers.

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Historically, most economists and policymakers have thought of generic drugs as a competition success story. When branded drugs lose patent protection, an influx of generics typically follows, capturing more than half of the market at less than half the branded equivalent's price (Scott Morton 1999; Goldman et al. 2011). In recent years, however, the prices of many generics have risen substantially. According to the complaint described below, many of these increases can be traced to a single precipitating event: In 2013, Teva Pharmaceuticals, the largest generic firm, hired NP, a marketing executive with especially strong industry relationships, and tasked her with "price increase implementation."¹ Over an 18-month period, industry participants allegedly exchanged thousands of calls and texts—alongside countless LinkedIn, Facebook, and WhatsApp messages and face-to-face conversations—with contacts at rival firms to coordinate the increases (*Connecticut et al. v. Teva Pharm. USA, Inc.* 2019, p. 322).² Following this period, prescription drug expenditures by governments, private insurers, and individuals rose sharply by billions of dollars.

In this paper, we measure the effect of price-fixing on market entry, estimate a structural model of generic drug competition, and use our estimates to assess counterfactual behavior and policies. We exploit detailed information on the cartel's internal operations, which were revealed in the complaint filed in May 2019. The complaint presents witness testimonies, private communications within and between rival firms, and internal documents, collected in the course of an extensive government investigation. It includes a list of the drugs (i.e., substance-delivery-release-strength combinations) for which NP fixed prices, the dates on which those prices were fixed, and the criteria NP used to select which prices to fix. It also reports NP's own numerical measure of the strength of her relationships with the sales and marketing executives at competing firms.

Information from the complaint is crucial to our research design, which compares cartelized and uncartelized drugs before and after collusion. Such comparisons typically raise concerns about unobservable differences across markets and time that relate to both the likelihood of collusion and outcomes of interest. However, information in the complaint reveals how NP selected markets, so we can observe the variables on which the selection depends. These variables are both stable over time and independent of the outcomes of interest. Also, the vast majority of the price increases were implemented within about a year of NP's first anniversary of joining Teva, mitigating concerns that they were timed to coincide with unobservable environmental changes. Moreover, observable outcomes in cartelized and uncartelized markets track extremely closely with each other leading up to collusion (but clearly diverge sharply afterward).

¹ We identify individuals using their initials. Their names are discoverable in court documents but irrelevant to our analysis.

² For purposes of our analysis, we have assumed that the allegations made in the May 10, 2019, version of the complaint filed by 44 States Attorneys General against 20 of the largest generic pharmaceutical manufacturers and 15 individuals alleging price-fixing are true. (The complaint is cited below.) We note that several defendants have settled the claims against them, and Teva settled a related case brought by the US Department of Justice, admitting that it engaged in illegal price-fixing and paying \$225 million in fines, agreeing to donate \$50 million in drugs, and selling a line of business. Teva also settled for \$420 million a securities lawsuit brought against it for failure to disclose its price-fixing activities to investors. Based on this assumption, throughout the paper, we use words such as "price-fixing," "collusion," "cartel," and "conspirators" to characterize certain behavior, economic arrangements, and individuals. The language we use reflects our subjective interpretation of statements made in the complaint. It does not reflect any finding of any court or admission of guilt by any entity or individual named in this paper.

Our sample consists of drugs manufactured by Teva in the quarter just prior to NP joining the firm. It includes all firms' versions of those drugs and spans 2008 to 2019. For example, Teva manufactures pravastatin sodium 100 mg immediate-release tablets in the first quarter of 2013, so this drug is in our sample. We measure quantity using prescriptions, which we obtain from IQVIA and Medicaid, and we measure point-of-sale price in dollars per prescription, which we obtain from Medicaid. Before manufacturing a drug, firms must file an Abbreviated New Drug Application (ANDA) and have it approved by the Food and Drug Administration (FDA). ANDA filings closely correspond to entry. We obtain the filings and their respective dates from the FDA.³

We find four key patterns in the data. First, prices rise sharply following cartel formation. Compared to uncarterelized markets, there are price increases that average about 50 percent in cartelized markets. Second, cartel formation is followed by significantly more entry, as measured by ANDA filings. Third, regulation delays entry by two to four years. As a result, a firm that filed an ANDA as early as 2013 might not be authorized to enter before, say, 2017. Fourth, the data suggest that entry by nonmembers exerts downward pressure on drug prices.

These facts inform our structural model, which emphasizes two main decisions that generic firms face. In the first stage, firms *without* regulatory approval file ANDAs if the sum of expected discounted profits associated with entering exceeds the sunk costs, given their expectations about future pricing behavior. Cartel members do not coordinate first-stage decisions (even though they may cooperate in the second stage).⁴ Entry is not immediate since the process of obtaining ANDA approval involves significant, stochastic delays.

In the second stage, firms *with* regulatory approval set prices. In uncarterelized markets, and in cartelized markets prior to cartel formation, firms set prices that maximize individual profits. A cartel forms when NP reaches an agreement to set prices that maximize the joint profits of the members, in which case nonmembers set prices that best respond. We assume the agreement is supported by trigger strategies, so realized prices depend on the history of play as well as incentive compatibility constraints (ICCs). The model highlights how cartel formation raises prices, which in turn increases incentives to enter. Moreover, it emphasizes that entry exerts downward pressure on price, given that not all entrants are cartel members.

Next, we take the model to the data. We estimate demand, use these estimates to recover marginal costs, and combine these results to forecast the profits that firms would earn for any hypothetical market structure and conduct. We use these forecasts in conjunction with the empirical distribution of entry delays to compute the value of entry, which equals the sum of discounted expected future profits. Firms enter if and only if the value of entry exceeds the sunk cost of doing so, so observed entry

³ All approval dates are reported by the agency, but only a subset of filing dates are. However, ANDA numbers are issued approximately sequentially, so filing dates can be inferred without meaningful error. See Supplemental Appendix A for a detailed description of the process.

⁴ Fershtman and Muller (1986) call this "semicollusive" behavior: Long-run choices are competitively chosen, but short-run choices may be cooperatively decided. As for why NP could not restrict entry by members, our conversations with industry participants suggest it entailed too much risk. At some firms, entry and pricing decisions are approved by different managers, meaning that restricting entry would require many more individuals be complicit in cartelization. For an illuminating case of strict market segmentation, see Clark et al. (2018). Note, though, that the asphalt cartel they study resorted to violence, which is presumably infeasible in our setting.

decisions map to the parameters that determine those costs. Since two types of firms (i.e., cartel members and nonmembers) make entry decisions, multiplicity is the rule rather than the exception. Thus, we rely only on the necessary conditions implied by Nash equilibrium, bounding the parameters of interest using moment *inequalities* (Tamer 2003; Pakes et al. 2015). These methods yield parameter estimates that imply accurate profit margins and sensible sunk costs of entry when compared to outside sources, such as audited financial statements or remarks by former FDA officials.

Our estimates reflect assumptions about conduct. To ensure the assumptions are consistent with the individual incentives of the cartel members, we introduce a dynamic games framework. Following Igami and Sugaya (2022), we measure each member's incentives to comply with the cartel agreement, plot the lower envelope over time across market structures and draws of the disturbances, and find that trigger strategies form an equilibrium. We then test whether the observed prices are consistent with our conduct assumptions, following Backus, Conlon, and Sinkinson (2021). Most notably, we find that collusive prices persist long after the government's investigation and ongoing prosecution. Presumably, this reflects tacit collusion, which would be consistent with statements by NP that imply that price-fixing in the generic drug industry is hard to initiate but easy to sustain. The data also reject a model in which nonmembers price cooperatively in favor of a model in which they best respond to cartel behavior, which means entrants can exert downward pressure on prices.

With estimates of the parameters governing demand, prices, and entry in hand, we simulate equilibria under different assumptions and policies. First, to assess the importance of observed entry on cartelized markets, we counterfactually prohibit firms from entering markets in which cartels have formed and recompute second-stage outcomes. Our most striking finding is that observed entry reduces profits earned by incumbent members by three-quarters, reflecting both lower prices and smaller market shares.

Second, to inform policymakers, we counterfactually reduce the regulatory costs and delays by up to \$600,000 and two years, respectively. These alternative policies are logical and important for several reasons. First, generic production requires long and expensive drug-specific government approvals. Firms must prepare and file an ANDA with the FDA, and the approval process can take years and cost millions of dollars. Second, these factors are within the government's control: The FDA has varied fees considerably over the past decade and experimented with various expedited approval programs. Former commissioner Scott Gottlieb has lamented both the cost and time associated with entry (Gottlieb 2016). Third and perhaps most important, regulatory costs and delays are basic features of many large industries. Examples range from air travel and power generation to defense contracting and retail banking. Hence, our broad conclusions are likely to be relevant beyond prescription drug markets. We find that reducing entry costs leads to significant entry but relatively modest consumer benefits, which are roughly offset by lost profits. In contrast, we find that reducing delays has a large effect on consumer surplus; the consumer compensating variation reaches \$1.5 billion.⁵

⁵These gains can be compared to potential costs of relaxing regulations. For generic drugs, we note that the active ingredients have already been deemed "safe and effective" by the FDA. Indeed, policymakers have moved to streamline the approval process with little concern about safety. However, for additional analysis of these trade-offs, see Grennan and Town (2020).

We find that entry can play a key role in disciplining cartels, with a first-order impact on consumer welfare. Nonetheless, the discipline it exerts may be incomplete and slow due to high costs and long delays, many of which can be attributed to regulation. We contribute to the growing body of empirical work on cartels by incorporating equilibrium entry behavior and post-entry pricing. While foundational work by Stigler (1964) names both cheating and entry as threats to collusive agreements, most early and influential formalizations of the cartel problem (e.g., Green and Porter 1984) study the former threat but rule the latter out. Harrington (1984) illustrates how the profitability of entry depends on post-entry pricing behavior.⁶ Fershtman and Pakes (2000) allow for entry by way of simulations, numerically solving Markov perfect equilibria that permit entry, exit, and quality differences across firms. Fershtman and Pakes (2000) report a strong industry relationship between collusion and subsequent entry, a relationship they state “has largely been ignored in the literature and has important implications for the welfare analysis of collusive behavior.”⁷

Empirically, Levenstein and Suslow (2006) note that entry can “undermine the best-laid collusive plans.” Collusion is often difficult to sustain, absent barriers to entry (Levenstein 1995; Symeonidis 2003; Vasconcelos 2004); cartels may take steps to limit entry (Levenstein and Suslow 2011; Harrington, Hüschelrath, and Laitenberger 2018) and yet may need or choose to accommodate some entrants (Scott Morton 1997; Podolny and Scott Morton 1999). In many recent empirical papers, authors have carefully chosen markets where it is reasonable to assume that entry is either unlikely to occur in the near term (Miller and Weinberg 2017) or that entry is determined by factors that are “outside” the model (Porter 1983; Byrne and de Roos 2019; Igami and Sugaya 2022).⁸ We model endogenous entry and subsequent pricing, which allows us to estimate the sunk costs and perform counterfactuals. To do so, we draw on work by Borenstein (1989); Bresnahan and Reiss (1991); Mazzeo (2002); Seim (2006); and the large, growing literature that followed. In contemporaneous work, Carrera (2022) takes a similar approach in his study of Chilean nitrate cartels. He finds that their profits were reduced by around 40 percent due to new entrants.

This paper also improves our understanding of prescription drug pricing (see, for example, Berndt, Conti, and Murphy 2018). Scott Morton (2000) shows that

⁶While incumbent firms may be unable to maintain collusion under the threat of entry when the anticipated response is discontinuing cooperation or accommodating new firms, if potential entrants expect the incumbent firms to respond aggressively, incumbents may be able to sustain collusion even absent high entry costs (Harrington 1989b).

⁷In particular, Fershtman and Pakes (2000) report, “The contrast between the entry states in the model that allows for collusion and the entry states in the model that does not is quite striking. The [relationships between entry and the state variables] simply *disappear* when collusion is not allowed” (emphasis in original). De Roos (2006) notes that the dynamics of entry that characterize the lysine cartel are not adequately addressed in the literature. Fershtman and Pakes (2000) emphasize the universally positive effect collusion has on entry; however, their statement contains a slight but especially interesting caveat due to work by Asker (2010), who studies bidding rings among stamp deals and shows how their arrangement induces *overbidding*, which deters entry.

⁸For example, Igami and Sugaya (2022) study vitamin cartels in the 1990s. In their setting, competition unexpectedly arose from technology change, which benefited fringe suppliers. Igami (2015) also considers the impact of fringe Vietnamese coffee producers. We build on their approach by modeling the sunk cost of entry. Elsewhere, Miller and Weinberg (2017) study the effect of the MillerCoors joint venture on the US beer market. In their setting, the emergence of another major brewer is very unlikely, at least in the near future, given that the leading brands have accumulated brand equity over decades of advertising, which insulates them from start-up competition. Moreover, while the “craft” segment has grown over the past three decades, Miller and Weinberg’s (2017) Table I shows the top two firms have nearly two-thirds of nationwide share. Entry barriers are not necessary, however, to restrict entry. For instance, Harrington (1989a) shows that sufficiently patient incumbents can credibly threaten entrants.

larger revenue markets, markets with more hospital sales, and products that treat chronic conditions attract more entry. At the same time, Ganapati and McKibbin (2023) show that entry—a traditional mechanism for reducing market power—has been limited in the United States in recent years. Mulligan (2021) argues the problem is burdensome regulation. We expand this literature by exploring the strategic incentives faced by generic manufacturers and the impact of these incentives on consumers. Cuddy (2020) models this market as a series of simultaneous procurement auctions, in order to study the equilibrium relationship between competition and prices. She estimates large damages from cartel behavior using a counterfactual auction model of competitive behavior. Similarly, Clark, Fabiilli, and Lasio (2022) estimate large damages using a reduced-form approach. We focus on the market forces that may serve to alleviate these harms.

The paper is organized as follows. Sections I and II describe the institutional setting and data, respectively. Section III reports patterns in the data that motivate our structural model. Sections IV and V describe and estimate the structural model. Section VI measures incentives to collude and tests the assumptions we maintain. Section VII simulates counterfactual outcomes under alternative assumptions and policies, and Section VIII concludes.

I. Industry Setting and Cartel Operations

A. The United States Generic Prescription Drug Industry

Generic drugs are a competition success story. When a branded drug loses exclusivity, generic entry drives prices down toward marginal cost. Generally, the first generic entrant will price its product slightly lower than the branded drug, and the second generic entrant will reduce the price to approximately 50 percent of the branded drug price. Conditional on having a large number of entrants, prices fall to around 20 percent of the price of the branded counterpart (Scott Morton 1999). For this reason, the market is the “most dynamic and cost-effective in the world” (Scott Morton and Boller 2017). Historically, barriers to entry have been relatively low. Under the Hatch-Waxman Act, firms can enter by filing an ANDA that shows that the active ingredient, delivery mechanism, strength, and dose of the generic drug are the same as the branded drug. The generic drug must be “bioequivalent” to the branded drug.

Generic drug manufacturers compete with each other to sell their products to wholesalers, distributors, and in some cases, directly to retail pharmacy chains, mail order and specialty pharmacies, hospital chains, and some health plans. Due to complex “cost-plus” reimbursement rules, higher wholesale prices may weakly benefit these market participants. In recent years, wholesale prices for some generic molecules have increased substantially. The increases have been attributed to both supply shortages and anticompetitive behavior (Cuddy 2020).

B. The Cartel

The cartel we study traces back to a single change in industry leadership: On April 22, 2013, NP joined Teva Pharmaceuticals. This event was special for several

reasons.⁹ First, in the years leading up to the cartel's formation, NP forged uniquely strong relationships. She worked as the Director of Global Generic Sourcing for Amerisource Bergen (ABC), one of the three major US drug distributors. The role led to "routine interaction with representatives from every major generic drug manufacturer" (*Connecticut et al. v. Teva Pharm. USA, Inc.* 2019, p. 158). Second, Teva was—and still is—the world's largest generic manufacturer. By early 2013, for example, it produced about one out of every three generic tablets and capsules. Third, NP's role as Director of Strategic Customer Marketing involved, in her own words, "price increase implementation" (*Connecticut et al. v. Teva Pharm. USA, Inc.* 2019, p. 158). The significance of this move was not lost on other industry leaders. At Taro Pharmaceuticals, another leading generic firm, the Vice President of Sales and Marketing emailed the COO just days after NP left ABC to say, "[NP] Going to Teva—Hush Hush for now" (*Connecticut et al. v. Teva Pharm. USA, Inc.* 2019, p. 159). Our sample includes only drugs that Teva historically produced. As the complaint states, Teva was uncooperative before NP joined.

The complaint states that just eight days after joining Teva, NP began identifying target markets. The process was highly structured. She started by assigning each generic firm an individual "quality" rating, which ranged from -3 to 3 , and corresponded to the strength of her relationships with their sales and marketing executives. Next, she combined these ratings into a drug-specific score, which was the most important element in the selection process. Finally, she factored in the number of firms in the market and other minor considerations. In an internally distributed 2014 document, she summarizes "Candidate Identification" as "Target 2–4 total players, where the quality of competitor is high" and "[where] Teva has majority share and quality of competitors is high" (*Connecticut et al. v. Teva Pharm. USA, Inc.* 2019, p. 217).¹⁰

The complaint states that the outcome of the process was a spreadsheet titled "Immediate PI File" (where PI stands for "price increase"). It contained a list of drugs for which NP expected to cooperatively raise prices. NP forwarded the list to supervisors on May 24, and Teva changed prices on July 3, preceded or followed closely by other firms. The magnitudes of the increases are especially noteworthy. Many commonly prescribed medications—drugs treating cancer, bacterial infections, arthritis pain, and high blood pressure, to name a few—doubled or more in price.

Fixing prices required considerable coordination, which in turn required frequent communication. Leading up to the increases, phone records reveal thousands of calls and messages between NP and her counterparts at firms producing the drugs for which she sought to raise prices. For instance, the day before an increase, NP exchanged 15 calls with 3 individuals. Interestingly, since perfectly synchronizing price changes is impossible, sharp increases by one firm often prompted its customers to approach

⁹While the Complaint focuses on drugs affected by NP, it alleges other segments of the industry were not entirely immune to coordinated behavior. These allegations do not affect our analysis. The Complaint describes antitrust violations in the other segments that are qualitatively and comparatively unimportant in our setting.

¹⁰The Complaint sets out NP's quality ratings, and we observe market structure, so we can confirm that NP used precisely these criteria. Firm quality is presented in Supplemental Appendix Table A.2, and average quality across markets is presented in Supplemental Appendix Figure A.1. She also mentions targeting "Exclusive items" (i.e., those where Teva is the only supplier). Our focus is collusion, so we exclude these cases from our sample. They are rare, so this restriction is without any meaningful loss of generality.

other producers, who were forced to decline the business. In some cases, firms outright declined, citing fictitious supply problems (*Connecticut et al. v. Teva Pharm. USA, Inc.* 2019, pp. 239 and 265), while in others they quoted inflated prices—a tactic NP called “fluff pricing” (*Connecticut et al. v. Teva Pharm. USA, Inc.* 2019, p. 146). This frustrated buyers such as Schnuck’s, a Midwest grocery chain, which felt “so insulted” by one of the egregious quotes (*Connecticut et al. v. Teva Pharm. USA, Inc.* 2019, pp. 146–47). Ultimately, NP coordinated five main rounds of price increases between July 2013 and January 2015 (*Connecticut et al. v. Teva Pharm. USA, Inc.* 2019, p. 3). Notably, the rollout would have been even more compressed had NP not taken maternity leave in the middle of it, which resulted in a six-month “hiatus” (*Connecticut et al. v. Teva Pharm. USA, Inc.* 2019, p. 212).

According to the Connecticut Attorney General (AG), “suspicious price increases of certain generic drugs” prompted the state in July 2014 to begin an investigation.¹¹ On May 10, 2019, 43 US states and territories filed a complaint, which was unsealed the following month. It alleges a “horizontal conspiracy” to fix prices for multiple generic drugs in violation of Section I of the Sherman Act. Alongside the states’ civil suits, the US Department of Justice brought criminal charges against several firms and individuals.

The complaint states that NP’s supervisor was “aware of the government investigations that had been commenced” by early 2015. She “told [NP] that the government was showing up on people’s doorsteps,” and “warned [NP] to be careful about communicating with competitors” (*Connecticut et al. v. Teva Pharm. USA, Inc.* 2019, p. 341). Again according to the complaint, news of the investigation ended the price increases (*Connecticut et al. v. Teva Pharm. USA, Inc.* 2019, p. 338). However, the news does not appear to have undermined existing agreements, which the firms may have maintained tacitly. This is consistent with NP’s assertions that generic drug price-fixing is hard to initiate but easy to sustain (*Connecticut et al. v. Teva Pharm. USA, Inc.* 2019, p. 160) as well as internal documents referencing cooperative arrangements years after the states launched their investigations (*Connecticut et al. v. Teva Pharm. USA, Inc.* 2019, p. 50). We address this issue directly in Section VI.

The complaint further suggests that cartel members—firms participating in the collusive scheme in any market—understood the value of sticking to the agreement. It is also worth noting what type of conduct is absent from the complaint and related filings. Despite thousands of pages of court documents recounting almost countless conversations about prices, no statements suggest that cartel members coordinated entry. In other words, it seems very unlikely that members made agreements not to enter each others’ markets. Consistent with that view, when entry by a cartel member into a cartelized market did occur, incumbent producers responded by accommodating rather than retaliating against them. Examples described in the complaint include ANDA launches by Aurobindo, Lupin, and Actavis (*Connecticut et al. v. Teva Pharm. USA, Inc.* 2019, pp. 74, 81–82, and 103–104, respectively). We test for

¹¹To be precise, the investigation traces back to a *New York Times* article titled “Rapid Price Increases for Some Generic Drugs Catch Users by Surprise” (Rosenthal 2014). A supervisor in the Connecticut AG Office’s unit of antitrust and fraud read and forwarded the article to a staff attorney, who subsequently issued subpoenas (Pazniokas 2019).

this behavior explicitly in Section VI. We further use our conduct tests to validate assumptions about expectations of post-entry pricing behavior.¹²

II. Data

A. Sources

The data come from several sources. The National Drug Code (NDC) Directory (US Food and Drug Administration 2022d) provides a current list of all prescription pharmaceutical products manufactured for sale in the United States. Each product has a unique NDC code, which identifies the substance-delivery-release-strength combination, the firm that produces it, and the ANDA that authorizes its production. The FDA updates the list daily. Using the Internet Archive, we take annual snapshots of the directory, which are the starting point for our panel dataset.

Medicaid State Drug Utilization Data (Centers for Medicare and Medicaid 2022b) is the primary source of price and quantity information. Each quarter, all states and the District of Columbia report the number of prescriptions filled by Medicaid enrollees and the corresponding expenditure. To compute price, we divide expenditure by quantity.¹³ We download national aggregate statistics, and we merge to the NDC Directory data, described above, at the NDC-quarter level. Since NDCs are specific to drug packaging (e.g., 1,000-count bottles, 14-count blister packages, etc.), firms are associated with multiple NDCs per drug. Thus, we sum over NDCs to get drug-firm-quarter observations.¹⁴

We also use quantity data from IQVIA (2022), a private provider whose National Prescription Audit covers 92 percent of US pharmacies. The IQVIA data provide detailed information on aggregate quantities at the drug-firm-quarter level but no information on price. We first use IQVIA data to scale up the quantities reported in State Drug Utilization Data, which covers only Medicaid, to US totals. We do this by computing the ratio of total IQVIA prescriptions to total Medicaid prescriptions, and we scale up by that factor. Second, we use IQVIA to check carefully whether patterns in Medicaid are representative of industry-wide purchasing behavior. Total purchases track extremely closely with one another over time, and their responses to cartel-induced price changes are almost identical (see Supplemental Appendix Figure A.2).

The complaint (*Connecticut et al. v. Teva Pharm. USA, Inc.* 2019), which is described in the previous section, provides information about cartel formation. The complaint lists the drugs for which NP allegedly fixed prices, the dates on which those prices were fixed, and NP's relationships. From this information, we construct

¹²Further evidence comes from Civica Rx, a start-up generic manufacturer owned by a consortium of hospitals. In a *New York Times* article titled "Fed Up with Drug Companies, Hospitals Decide to Start Their Own" (Abelson and Thomas 2018), the firm stated that when it enters the market, it expected incumbents to respond by "quickly dropping the price of the drugs in question." Further, the firm stated that were it to exit subsequently, it expected incumbents to respond by "raising them again later."

¹³We account for the fact that each extended release dose commonly substitutes for two evenly spaced immediate-release doses. We manually change two prices.

¹⁴Publicly available price data are also available from National Average Drug Acquisition Cost (NADAC) surveys (Centers for Medicare and Medicaid 2022a), but these were first administered in 2013, so this source does not cover our analysis period.

an indicator variable for whether a cartel was ever formed in a particular drug market (i.e., market for a particular substance-delivery-release-strength). Within each drug market where a cartel is formed, we also construct an indicator for whether a particular period precedes or follows cartel formation. Finally, we construct an indicator variable for cartel membership.

We then merge in ANDA filing and approval dates. To obtain filing dates, we scrape US Food and Drug Administration (2022b) for all available ANDA approval letters, most of which state the date the application was filed. ANDA numbers are largely assigned in chronological order. We were able to parse filing dates for 8,185 ANDAs from PDFs. We supplement our parsed dates using information from Feldman et al. (2016) and infer the rest using the FDA's numbering system (e.g., if ANDA 70001 was filed on January 1 and ANDA 70015 was filed on January 10, we infer that ANDAs 70002–700014 were filed between January 1 and January 10). To fill in missing filing dates, we regress ANDA numbers on filing dates and then assess fit. To minimize measurement error, we aggregate to the quarterly level. See Supplemental Appendix Figure A.3 and the associated discussion, which shows that we measure ANDA filing dates without meaningful error. We obtain approval dates from the Orange Book (US Food and Drug Administration 2022a). We also merge each product's substance to a drug class (Norwegian Institute of Public Health 2022).

Our sample covers 2008 to 2019 and consists of all drugs that Teva manufactured in the quarter prior to NP joining the firm, regardless of manufacturer, subject to some exceptions. We omit drugs that lose exclusivity during the sample period (US Food and Drug Administration 2022c), as they exhibit drastic price declines that are unrelated to cartel formation and ruin comparability across units. We also omit 11 substance-delivery-release combinations for which NP raised prices twice. Last, we drop 38 injectable drugs: Injectables occupy a very different segment of the industry, require radically different production methods, have short shelf lives, are mostly manufactured by different producers than those we study, and most importantly, are never mentioned in the complaint. The sample construction is described in detail in Supplemental Appendix Table A.1.

B. Summary Statistics

Table 1 summarizes the data at the drug-year level. There are 4,992 observations spread over 416 drugs and 12 years. The average number of prescriptions filled is 1.8 million, and the average price is \$32. Average yearly expenditures are \$30 million, or about \$250 billion over the full panel. The mean number of manufacturers in the market is 4.15, while the mean number of ANDA filings is 0.21, highlighting that entry into generic drug markets is typically rare.¹⁵ Most ANDA filings can be attributed to cartel “nonmembers” (i.e., firms that are not named in the complaint). The group accounts for about 60 percent of entry over the full sample and closer to 67 percent in the periods after NP joined Teva, regardless of whether we consider

¹⁵The exception is the period immediately following the loss of exclusivity of the reference branded drug. Using data from an earlier period, Scott Morton (1999) shows that 80 percent of entry happens within six months after the initial generic approval.

TABLE 1—SUMMARY STATISTICS

	Count	Mean	SD	Minimum	Maximum
Price	4,992	31.66	40.42	2.32	628.53
Quantity (in thousands)	4,992	1,819.33	3,572.70	0.15	29,571.61
Expenditure (in millions)	4,992	29.90	110.08	0.01	4,188.54
Number of firms	4,992	4.15	1.93	1.00	12.00
Number of ANDA filings	4,992	0.21	0.54	0.00	5.00
Cartelized drug	4,992	0.27	0.44	0.00	1.00
Cartelized drug $\times$ Post cartel formation	4,992	0.14	0.34	0.00	1.00

Notes: The unit of observation is a drug-quarter. Price refers to dollars per prescription. Quantity is measured in thousands of prescriptions. Expenditure is measured in millions of dollars. “Cartelized drug” is an indicator for whether a cartel formed in the drug market at any point in the sample. “Cartel drug $\times$ Post cartel formation” is the interaction of “Cartelized drug” and an indicator for periods following cartel formation (i.e., the indicator is “on” if the cartel for a particular drug formed in April 2013 and the observation summarizes 2013:III or later).

all drug markets or just those in which cartels are eventually formed (not shown). Cartels are formed in the markets for 113 drugs, or 27 percent of the sample.¹⁶

III. Descriptive Analysis

In this section, we describe the patterns in the data that motivate the structural model.

A. Price Changes Following Cartel Formation

We first ask whether and to what extent prices increase following cartel formation. To answer this, we plot the average log prices of cartelized and uncartelized drugs against calendar time in quarters. We normalize each price series to zero in the period just prior to the one in which NP joins Teva, and we mark that period with a vertical red line.

Figure 1 reports the result. Four aspects of the graph stand out. First, the prices of cartelized and uncartelized drugs exhibit very similar trends in the periods leading up to Teva hiring NP. Second, within the pre-event trends, quarter-to-quarter innovations in the price series are correlated. Third, the prices of uncartelized drugs trend smoothly through Teva’s hiring of NP, suggesting that the generic markets were not affected by any major industry-wide shocks (aside from cartel formation). Fourth, the prices of cartelized drugs rise sharply after NP joins Teva; they increase about 50 percent by the end of 2014.

We report supplementary results in Supplemental Appendix B. In Supplemental Appendix Figure B.3, we show that the relationship between the “quality” scores assigned by NP and the likelihood of cartel formation is so tight that we can produce a graph similar to Figure 1 by comparing “high-quality” and “low-quality” markets (rather than cartelized and uncartelized ones). In Supplemental Appendix Figure B.1, we provide a more precise sense of timing and magnitude by comparing

¹⁶NP selected markets where she has close relationships with substantively all of the other firms. The probability this occurs falls with the number of firms in the market, so it also falls with market size. Thus, cartelized markets should be somewhat smaller and have fewer firms. Supplemental Appendix Table A.3 confirms this prediction.

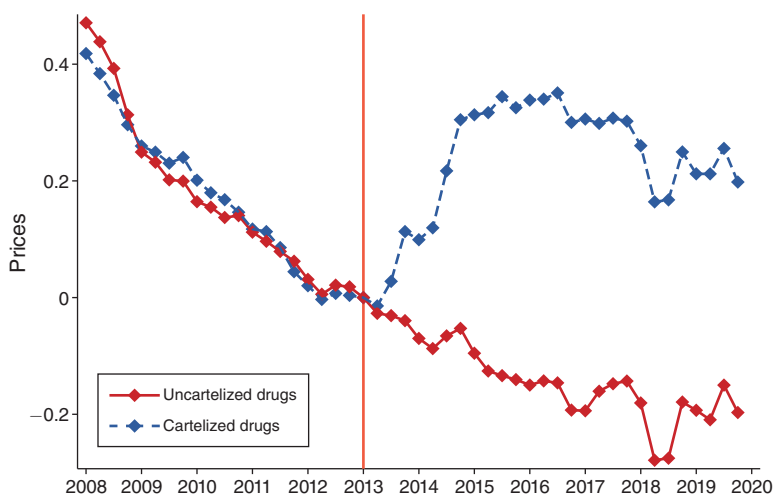


FIGURE 1. PRICES RISE SHARPLY FOLLOWING CARTEL FORMATION

Notes: This figure plots the average log price of cartelized and uncartelized drugs on the y-axis against calendar quarter on the x-axis. The vertical red line corresponds to the first quarter of 2013, the period in which NP joined Teva. Prices are normalized to zero in that quarter.

prices before and after cartel formation, conditional on drug and year fixed effects. In Supplemental Appendix Figure B.2, we produce an analogous graph, but we rely on an approach suggested by Sun and Abraham (2021) rather than on two-way fixed effects. Their approach accounts for potential contamination of leading and lagging coefficients but produces nearly identical results, indicating that our findings are not driven by variation in when the markets were cartelized. Finally, in Supplemental Appendix Figure B.4, we show that very different patterns emerge when we restrict attention to markets in which Teva is a monopolist.

B. Entry Following Cartel Formation

We next examine entry. Before a firm launches a generic product, it must file an ANDA and then obtain FDA approval. We begin by studying the first step in the process captured in our data, ANDA filings. Since ANDAs apply to all strengths within a substance-delivery-release combination, we measure entry at this level throughout the paper. To improve legibility and simplify estimation and counterfactual exercises, we also aggregate outcomes to the annual rather than quarterly level for the remainder of the paper. We plot the average number of ANDA filings in cartelized and uncartelized markets against calendar time in years. We normalize each series to zero in 2013, the year NP joins Teva, and mark that period with a vertical red line.

Panel A of Figure 2 reports the result. Measured by ANDA filings, entry into cartelized and uncartelized drug markets tracks closely with one another prior to NP joining Teva, and entry into uncartelized markets trends smoothly through this

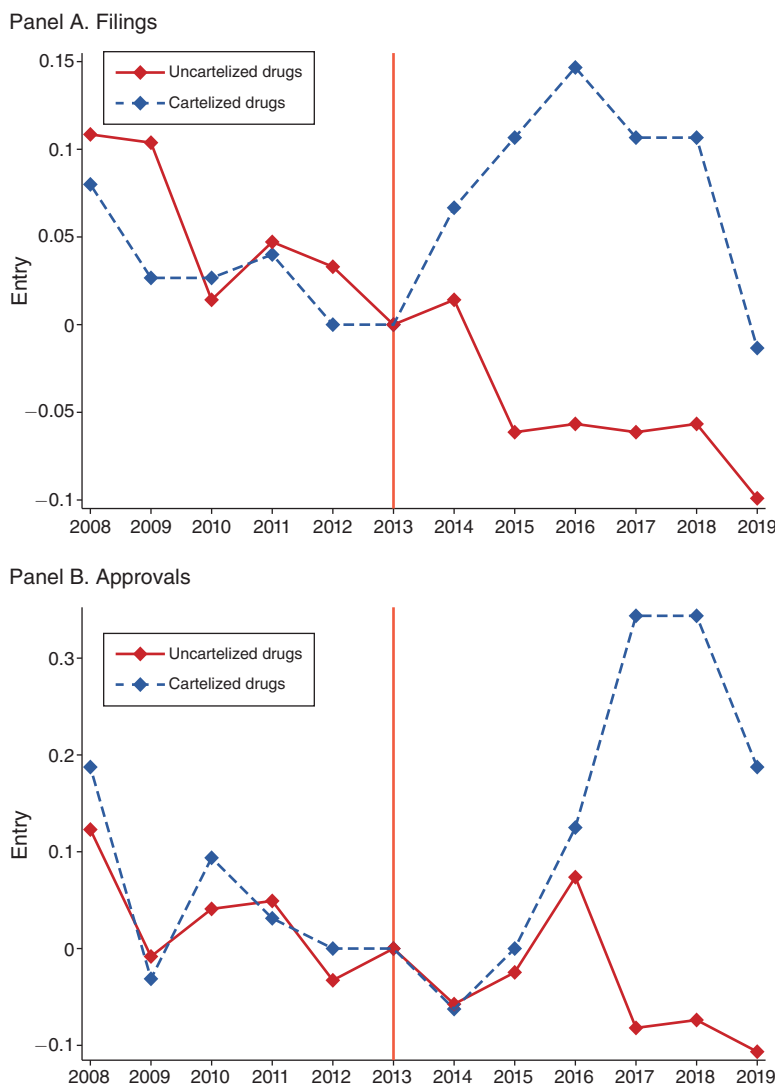


FIGURE 2. ANDA FILINGS RISE AFTER CARTEL FORMATION, BUT APPROVALS ARE DELAYED

Notes: This figure plots average number of entrants per substance-delivery-release-combination on the y-axis against calendar year on the x-axis. Panel A measures entry using ANDA filings, whereas panel B measures it using ANDA approvals. The vertical red line corresponds to the first quarter of 2013, the period in which NP joined Teva. The number of entrants are normalized to zero in that quarter.

event. Yet, as with prices, ANDA filings rise sharply in cartelized markets in the periods immediately following NP's hiring.¹⁷

¹⁷For completeness, we also plot filings and entry in event time relative to cartel formation (see Supplemental Appendix Figure B.6). Infrequently, producers reenter markets using dormant ANDAs (i.e., approved filings that are no longer associated with production but once were). We examine reentry in Supplemental Appendix B. Briefly, we find that reentry spikes in the year immediately following collusion. The pattern is consistent with the cartel turning some unprofitable markets profitable once again, and it squares with the fact that applications already approved can enter without delay.

ANDA filings do not permit immediate entry. The FDA must review and approve the ANDA, which involves delays. To capture effective entry, we repeat the process used to produce panel A but replace ANDA filings with ANDA approvals. Panel B of Figure 2 reports the result. Whereas ANDA filings in cartelized markets increase in 2014, ANDA approvals do not rise until 2017, consistent with regulatory delays of roughly three years.

For a more complete picture of delays, we plot the full distribution of delays measured in years. Supplemental Appendix Figure B.8 reports the result. The median time between ANDA filing and approval is three years, consistent with Figure 2. Notably, the distribution of total delays—the time between the formation of the cartel and the start of production—is shifted right by 1.5 years, suggesting ANDA preparation and/or factory setup also delay entry.

C. Post-entry Price Changes

The negative relationship between entry and prices is better established in US drug markets than perhaps any other industry (Goldman et al. 2011), but the situation may be complicated by collusion in our setting. Here, we present additional data patterns that are consistent with a relationship between entry and prices.

First, to compare markets where entry is more or less likely, we exploit variation in market size, following a large body of established work (see, e.g., Berry and Waldfogel 1999 and the work cited therein). Market size shifts the probability of entry but is less likely to be correlated with unobserved factors that influence prices.¹⁸ In Supplemental Appendix Figure B.9, we confirm that, following cartel formation, larger markets attract more entrants, echoing cross-sectional results reported in Scott Morton (1999). Then, in panel A of Figure 3, we compare prices in large and small cartelized markets over time, using prices of uncartelized drugs to control for any time-varying factors affecting the entire industry. To produce this, we estimate

$$(1) \quad y_{dt} = \sum_{\tau=-21}^{24} \beta^{\tau} x_{dt}^{\tau} + a_d + b_t + e_{dt},$$

where y_{dt} denotes the average log price of drug d in quarter t ; a_d and b_t denote drug and quarter fixed effects, respectively; and x_{dt}^{τ} denotes an indicator variable that equals one if and only if d is a cartelized drug and t is τ periods from cartel formation. Given the timing of cartel formation in relation to our sample period, we restrict attention to $\tau \in [-21, 24]$, and we set $\beta^{-1} = 0$ to facilitate comparisons to the period just prior to cartel formation. We first estimate equation (1) on uncartelized and small cartelized markets and plot the results along the dashed blue line.

¹⁸Cross-sectional differences in entry are plainly endogenous. Nonetheless, entry is associated with lower prices in cartelized markets. That is, prices in cartelized markets with and without entry rise sharply after cartel formation and peak around 18 months later, but they follow qualitatively different paths thereafter. Measured from their peaks to their values at the end of the sample, average prices in markets with entry fall 39 percent, whereas average prices in markets without entry are stable (i.e., they plateau and do not move more than about 1 percent in either direction).

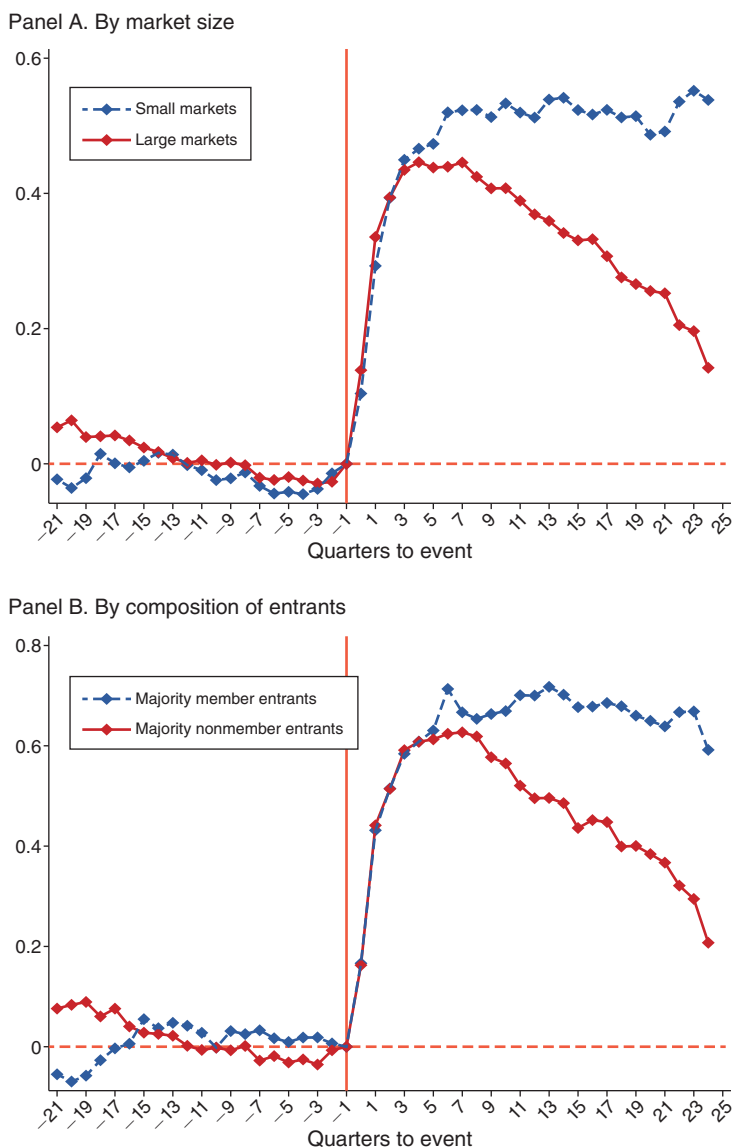


FIGURE 3. PRICES AFTER CARTEL FORMATION

Notes: This figure plots estimates of β^τ , which are derived from equation (1), against x^τ , which represents event time in quarters. For panel A, we proxy for market size by computing the number of prescriptions filled in the period just prior to NP joining Teva, and we distinguish between large and small cartelized markets based on how this figure relates to the median value. For panel B, we calculate the fraction of entrants that arrive as a result of ANDAs filed after NP joins Teva and identify markets in which a majority of those firms are members and nonmembers, respectively.

We then reestimate equation (1) on uncarterized and large cartelized markets and plot those coefficients along the solid red line.

Prices in large cartelized markets exhibit very similar pre-event trends as those in small cartelized markets. After cartel formation, both series rise sharply and reach similar peaks. After that point, the series diverge. Prices in large markets decline

throughout the remainder of the panel, while prices in small markets remain completely stable. Readers should obviously interpret this result cautiously: We cannot, for example, rule out the possibility that this divergence reflects, in part, an effort to reduce antitrust risk.¹⁹

Second, we expect that nonmembers drive the entry-price relationship. According to the complaint, NP repeatedly coordinated pricing decisions by member entrants to “avoid competition” (see, e.g., *Connecticut et al. v. Teva Pharm. USA, Inc.* 2019, p. 107). Yet nowhere in the over 450-page document do the states allege NP coordinated prices set by nonmember entrants (or nonmember incumbents, for that matter). To make comparisons based on the cartel affiliation of the entrants, we identify firms whose ANDAs were filed after NP arrived at Teva and compute the fraction that are members in each cartelized market. In panel B of Figure 3, we compare the prices of cartelized drugs for which the majority of entrants are members with the prices of cartelized drugs for which the majority are nonmembers, using the prices of uncartelized drugs as controls. We then follow exactly the same procedure that we use to produce panel A.

Again, we find that the two price series exhibit similar pre-event trends and reach similar peak prices about one year after cartel formation but diverge afterward. Cartelized markets in which most entrants are nonmembers experience declining prices, whereas ones in which most entrants are members experience very stable prices. These patterns are consistent with claims that member entrants comply with existing cartel agreements, while nonmembers do not.²⁰ Ultimately, incentives to enter depend on post-entry conduct. To address these issues, we turn to a structural model of entry, pricing, and demand.

IV. Model

In this section, we present a model of generic prescription drug supply and demand. For each drug in each period, there are two stages. In the first stage, firms that would like to enter but do not have regulatory approval file an ANDA. The FDA approves these filings after a delay of uncertain duration. In the second stage, firms that have regulatory approval manufacture and sell the drug to buyers. To ease exposition, we describe the model in reverse chronological order, starting with demand.

A. Stage II: Demand

Patients are the final consumers of generic prescription medications but are typically not the ones to decide which firm’s product to purchase. Instead, buyers are intermediaries such as wholesalers (e.g., Cardinal Health), group purchasing organizations (i.e., cooperatives formed to source drugs), and large retail chains (e.g.,

¹⁹On the one hand, high prices for popular drugs may generate headlines, in which case members may reduce prices in large markets in an attempt to avoid prosecution. On the other hand, when we trace the investigation back to its origin, obtain the list of drugs whose price changes triggered the inquiry, and compare them to all products manufactured in our sample, their sizes are similarly distributed (see Supplemental Appendix Figure B.10 for details.) Moreover, despite carefully describing how members allegedly responded to the states’ investigation (e.g., by deleting text messages), no statements come close to suggesting that they adjusted prices in response.

²⁰In Supplemental Appendix Figure B.11, we show that cartel formation affected entry among members and nonmembers similarly.

CVS). The buying process resembles a procurement auction. For every drug in every period, each buyer submits a request for proposals (RFP), to which any firm with an approved ANDA may respond. Next, the buyer assigns each proposal a score, which is the payoff associated with selecting that proposal's product. Buyers maximize payoffs by selecting the highest-scoring proposal or an outside option.

Scores depend on price and nonprice attributes. For instance, Cardinal Health states that it values not only low prices but also short lead times, frequent product availability reports, infrequent recalls, accurate invoicing, high historical fill rates, and high-quality, on-time deliveries (Cardinal Health 2020).²¹ Since nonprice attributes mostly reflect large, historic investments or past operating performance, we assume these aspects of the proposal are fixed when firms submit proposals.

To formalize the process, we index firms by f , buyers by i , drugs by d , and periods by t . The payoff associated with buyer i selecting the product made by f is given by

$$(2) \quad v_{idft} = \lambda_d + \lambda_t + \xi_{dft} + \zeta_{idt} + (1 - \sigma) \epsilon_{idft} + \alpha_d p_{idft}.$$

λ_d and λ_t capture differences in the prevalence of medical conditions and the popularity of treatment regimens that vary over drugs and periods, respectively. ξ_{dft} , ϵ_{idft} , and ζ_{idt} are i.i.d. disturbances.²² We assume ϵ_{idft} is distributed Type 1 extreme value and that ζ_{idt} is distributed such that $[\zeta_{idt} + (1 - \sigma) \epsilon_{idft}]$ is also distributed Type 1 extreme value. α_d measures the sensitivity of the payoff to price, denoted by p_{idft} , and may vary by drug. We assume firms know ξ but not ζ or ϵ when they choose prices, as is common in the literature. Without knowledge of how buyers idiosyncratically value products, firms set a single price for each product, meaning p_{idft} can be written as p_{dft} . Since buyers rank alternatives, the mean payoff assigned to the outside option is normalized to zero without loss of generality, so $v_{id0t} = \epsilon_{id0t}$. In our setting, this choice entails drawing down existing stock; aggressively optimizing inventory across warehouses and stores; and intermittently stocking out at certain locations, a common occurrence during our sample period (see, e.g., US Food and Drug Administration 2021).

In the complaint, state attorneys name 35 different buyers. These consist of distributors (Amerisource Bergen, Cardinal, McKesson, HD Smith, and Morris and Dickson), group purchasing organizations (e.g., Viziant, MMCAP, and Econdisc), insurers (e.g., Humana, NC Mutual, Cigna), supermarkets (e.g., Walmart, Target, Giant Eagle, Ahold), and pharmacies (CVS, Walgreens, Rite Aid, and Kinney Drug). Thus, we assume there are a large number of similarly sized buyers, which is a reasonable approximation of the actual “downstream” market structure. Under these conditions, we can treat the collection of procurement scoring auctions as a nested logit demand system (Einav 2003; Miller 2014; Miller and Sheu 2021). See Supplemental Appendix E for the proof.

²¹ See also the RFP template published by the Minnesota Multistate Contract Alliance for Pharmacy (MMCAP), which is available at <https://www.bidnet.com/bneattachments/489130294.pdf>.

²² Within a given drug-period, ξ captures product-specific shocks, such as manufacturing defects that temporarily reduce quality. For instance, certain lots of Aurobindo's valsartan were contaminated with the carcinogen N-Nitrosodiethylamine (NDEA). The lots were recalled, but news of the incident temporarily reduced demand for the firm's version of the drug. ϵ captures idiosyncratic buyer-product-specific shocks. For example, distributors including Cardinal value the ability to “backhaul” shipments, which ultimately depends on where its distribution facilities are located in relation to where the firms warehouse their product. ζ allows the score of “inside” products to be correlated with one another.

The market share of f in d at t is given by

$$(3) \quad s_{dft} = \frac{e^{\lambda_d + \alpha_d p_{dft} + \xi_{dft}}}{\Lambda^\sigma (1 + \Lambda)^{1-\sigma}},$$

where $\Lambda = \sum_{f' \in \mathcal{F}_{dt}} e^{\lambda_d + \alpha_d p_{dft'} + \xi_{dft'}}$ and $\mathcal{F}_{dt}$ denotes the set of firms producing d at t . The market share of the outside good, s_{d0} , is obtained by replacing the numerator in equation (3) with one and the denominator with $(1 + \Lambda)^{1-\sigma}$.

B. Stage II: Pricing

Profit is given by

$$(4) \quad \pi_{dft} = (p_{dft} - mc_{dft}) s_{dft} A_d,$$

where mc_{dft} denotes marginal cost and A_d denotes market size.²³ We parameterize marginal cost such that

$$(5) \quad \ln(mc_{dft}) = \gamma_d + \gamma_t + \omega_{dft},$$

where ω_{dft} is an i.i.d. disturbance.

In uncarterized markets, and in cartelized markets prior to cartel formation, we assume that all firms set *competitive* prices, which maximize individual profits. We denote competitive prices by p_{dft}^B and corresponding profits by π_{dft}^B . Alternatively, when a cartel forms, we assume that NP reaches an agreement with other members to set *collusive* prices, which maximize cartel members' joint profit. Based on facts presented in the complaint and patterns in the data, we also assume NP does not approach nonmembers to discuss the agreement, so each member best responds to the cartel. When members maximize their joint profit and nonmembers best respond, we denote each member's price by $p_{dft}^{C,M}$ and each nonmember's price by $p_{dft}^{C,N}$, and we denote the corresponding profits they earn by $\pi_{dft}^{C,M}$ and $\pi_{dft}^{C,N}$, respectively.

Merely reaching an agreement does not guarantee compliance. Absent other considerations, not complying is always profitable for members, so the agreement must stipulate punishment. Given the effort required to coordinate behavior, the legal risk involved with price-fixing, anger that would arise from betrayal, and subsequent loss of trust, it is likely that a single intentional deviation would not only undermine the cartel but also preclude its members from forming a new one. Thus, in each cartelized market, we assume cooperation among cartel members is supported by trigger strategies. All members agree that in every future period, each member will set collusive prices so long as every member has set collusive prices since cartel formation. They also agree that if any member does not comply, then all members will set competitive prices forever after. In this sense, noncompliance "triggers" reversion to Nash-Bertrand. In doing so, we abstract away from

²³ We use IQVIA data to compute the total number of prescriptions per year per drug. We compute the maximum value for each drug over the sample and set market size equal to 50 percent above this figure.

multimarket contact, that is, the possibility that firms punish noncompliance in one market by charging punitive prices in another.

Under trigger strategies, prices depend on the beliefs of firms and the history of play. In Section VI, we formalize the dynamic game, measure the incentives to comply with the cartel agreement at every point in time, report that trigger strategies form an equilibrium, and explicitly evaluate conduct. For the remainder of this section and the entirety of the next, we (temporarily) assume that once a cartel forms, it is in each member's individual self-interest to set collusive prices. Doing so greatly eases exposition and economizes on notation.

C. Stage I: Entry

In the first stage of the period in which a cartel forms, we assume that firms without an approved ANDA have a one-time opportunity to file one. Filing an ANDA allows a firm to enter after a delay of uncertain duration, denoted by D . For tractability, we assume that other market structure changes are determined outside the model. Since entry into mature generic drug markets is otherwise rare, and since exit is usually precipitated by supply disruptions outside a firm's control, this assumption greatly reduces the computational burden but is unlikely to have a material impact on our conclusions.

We assume that firms know the distributions of ξ , ω , and D and that they have rational expectations about future realizations of these random variables. We denote the cumulative distribution functions of ξ and ω by F_ξ and F_ω , respectively, and the cumulative mass function of D by F_D . Experienced firms learn about these objects over time, and nascent firms can easily estimate them from data similar to ours. We also assume that firms know how λ_t , γ_t , and the numbers of member and nonmember incumbents will evolve over time, and we assume that these objects are fixed at their end-of-sample-values. For mature generic drugs, market-wide demand and cost shocks are rare, exit is infrequent, and reentry is equally infrequent (provided at least four years have passed since cartel formation).

Since ANDAs apply to all strengths within a substance-delivery-release combination, our analysis of entry occurs at that level. We index substance-delivery-release combinations by j and denote the set of d corresponding to j by $\mathcal{D}_j$. The value of entering j equals the value of entering each drug market associated with it and depends on the number of member and nonmember entrants, denoted by χ_M and χ_N , respectively. For a member, the value of entering j is given by

$$(6) \quad VE_j^M(\chi^M, \chi^N) = \sum_{d \in \mathcal{D}_j} VE_d^M(\chi^M, \chi^N),$$

where

$$(7) \quad VE_d^M(\chi^M, \chi^N) = \sum_{t=1}^{\infty} \delta^t F_D(t) \sum_{e_M=0}^{\chi_M-1} \sum_{e_N=0}^{\chi_N} \left[\rho(\chi_M - 1, e_M, t; F_D) \rho(\chi_N, e_N, t) \right. \\ \left. \times \int_{\xi} \int_{\omega} \pi_{d,f,t_d+t}^{C,M} (M_{d,t_d+t} + e_M + 1, \right. \\ \left. N_{d,t_d+t} + e_N, \xi_{d,t_d+t}, \omega_{d,t_d+t}) dF_{\xi} dF_{\omega} \right].$$

δ denotes the discount factor. $M_{d,t}$ and $N_{d,t}$ denote the number of member and non-member incumbents in d at t , respectively. t_d denotes the period in which the cartel is formed. $\rho(\chi_M - 1, e_M, t; F_D)$ is the probability that e_M other cartel member entrants are active t periods after the cartel is formed, while $\rho(\chi_N, e_N, t; F_D)$ is an analogous probability for nonmember entrants. Both are binomial expansions given by

$$(8) \quad \rho(a, b, t; F_D) = \frac{a!}{(a-b)!b!} F_D(t)^b [1 - F_D(t)]^{a-b}.$$

The final term in equation (7), $\int_{\xi} \int_{\omega} \pi(\cdot) dF_{\xi} dF_{\omega}$, represents the expected variable profits earned by the potential entrant, assuming it is active at $t_d + t$. For a non-member, the value of entry is denoted by $VE_j^N(\chi^M, \chi^N)$ and given by an expression analogous to the one that appears on the right-hand side of equation (6). The payoff to a potential entrant that does not file an ANDA is normalized to zero regardless of cartel membership.

Potential entrants weigh the value of entry against expected sunk costs. We denote the sunk cost for f in j by θ_{jf} , its information set by $\mathcal{I}_{jf}$, and its expectation operator as $\mathbb{E}$. Cartel member f files an ANDA for j if and only if

$$(9) \quad VE_j^M - \mathbb{E}[\theta_{jf} | \mathcal{I}_{jf}] \geq 0.$$

Nonmember f' follows an analogous rule, which depends on VE_j^N instead of VE_j^M and $\mathbb{E}[\theta_{jf'} | \mathcal{I}_{jf'}]$ instead of $\mathbb{E}[\theta_{jf} | \mathcal{I}_{jf}]$. Entry decisions form a simultaneous move Nash equilibrium. More than one may exist.

V. Estimation

In this section, we describe the methods we employ to estimate the structural parameters. We also report the resulting estimates.

A. Stage II: Demand

To estimate the demand system, we rely on the market share inversion proposed by Berry (1994) and estimate

$$(10) \quad \ln(s_{dfi}) - \ln(s_{d0t}) = \lambda_d + \lambda_t + \alpha_d p_{dfi} + \sigma \ln(s_{dfi|dgt}) + \xi_{dfi},$$

where $s_{dfi|dgt}$ denotes f 's share of the inside good, which equals $s_{dfi} / (1 - s_{d0t})$. The left-hand side of equation (10) is a straightforward transformation of the data, and the right-hand side is linear in parameters to be estimated— λ_d , λ_t , σ , and α_d —and an error term. In our initial specification, we restrict $\alpha_d \equiv \bar{\alpha}$, but we partially relax this restriction below.

Since firms know ξ when they set p , prices are endogenously determined in equation (10), so OLS estimates of $\bar{\alpha}$ will be biased. We estimate the parameter using price variation induced by the cartel. Our price instrument equals the product of a dummy for cartelized drug markets and a dummy for $t \geq t_d$. The intuition behind the identification strategy is transparent. Following cartel formation, the price and

TABLE 2—DEMAND ESTIMATES

Variables	IV (1)	IV (2)	IV (3)	IV (4)
log of inside share	0.56 (0.16)	0.56 (0.16)	0.73 (0.30)	0.73 (0.30)
Price	−0.11 (0.031)	−0.11 (0.031)	−0.21 (0.10)	−0.21 (0.10)
Price $\times$ β -blocker			0.16 (0.089)	0.16 (0.089)
Price $\times$ anticonvulsant			0.20 (0.095)	0.20 (0.094)
Indicator for authorized generic		−0.043 (0.16)		−0.068 (0.31)
Observations	19,299	19,299	19,299	19,299
Drug fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Number of drugs	415	415	415	415

Notes: The unit of observation is a drug-year-firm. Standard errors are clustered at the drug level.

consumption of cartelized drugs rise and fall, respectively, compared to uncartelized drugs. The relative changes reflect sensitivity to price, which maps to the $\bar{\alpha}$. Given the origin of the cartel, the selection of drugs, and the timing, it is reasonable to assume that values of the instrument are uncorrelated with drug-firm-year-specific demand shocks.²⁴ In nested logit models, an additional right-hand-side term, $s_{dft|dgt}$, is also endogenously determined. Our instrument counts the number of products in the market, so we estimate σ using share changes following entry and exit.²⁵

Table 2 reports the demand estimates. Column 1 corresponds exactly to equation (10). We allow for a separate “nest” for the outside good and find that preferences are correlated within the “nest” of inside goods, with $\sigma = 0.56$. As expected, we also find buyers dislike higher prices, with $\bar{\alpha} = -0.11$. Standard errors are clustered at the drug level, and both coefficients are significant at the 1 percent level. In column 2, we assess the importance of accounting for “authorized generic” products. Authorized generics are generic versions of drugs that are manufactured by the patent owner, which are often introduced around the point at which it loses exclusivity. Throughout our analysis, we do not distinguish these products from “ordinary” generics. To ensure this distinction is empirically unimportant, we append equation (10) to include a dummy for authorized generics. Reassuringly, estimates of σ and α are entirely unchanged, and the coefficient on the authorized generic dummy is small and not significant.

We observe that two drug classes experience larger-than-expected price changes at the onset of collusion (as shown in Supplemental Appendix Figure C.1) and witness more subsequent entry than one would predict on the basis of market size

²⁴To focus on periods proximate to cartel formation, we restrict attention to a five-year window around cartel formation, and to avoid complications arising from midyear cartel formation, we omit $t = t_d$.

²⁵The instrument equals $M_{dt} + N_{dt}$. In the presence of drug-specific dummy variables, which all specifications include, the instrument captures period-to-period changes in the number of active firms. Though the parameters are jointly determined, one can build intuition around identification by considering exit, which reduces the value of the instrument. If the existing firm cedes a large share to other drugs instead of the outside option, this implies σ is relatively large.

alone. Both patterns are consistent with less elastic buyers. Thus, to accurately represent demand and correctly forecast entry incentives, we allow α_d to vary based on whether the drug is a beta-blocker, anticonvulsant, or other generic. That is, we replace $\bar{\alpha}$ on the right-hand side of equation (10) with $\alpha_0 + \alpha_1 \mathbf{1}_{\{\beta\text{-blocker}_d\}} + \alpha_2 \mathbf{1}_{\{\text{antiepileptic}_d\}}$, where $\mathbf{1}_{\{\cdot\}}$ denotes an indicator variable. Allowing α to vary more flexibly (e.g., by drug class) may be desirable but generates noisy estimates.

Column 3 of Table 2 reports the result. We find that $\alpha_1 = 0.16$ and $\alpha_2 = 0.20$, consistent with less elastic buyers in the separately named drug classes. As expected, we find a higher baseline price coefficient (in absolute value terms), with $\alpha_0 = -0.21$. We also estimate a stronger correlation of products within the nest of inside goods. All estimates are significant at the 5 percent level except α_1 , which is significant at the 10 percent level.²⁶

B. Stage II: Marginal Costs

To recover marginal cost, we restrict attention to these unambiguously competitive markets. Specifically, we isolate observations from (i) markets that were never cartelized and (ii) years prior to cartelization in markets that were cartelized. For each observation, we take the derivative of profit with respect to price and solve for marginal cost, which yields

$$(11) \quad mc_{dft} = p_{dft} - \left(\frac{\partial s_{dft}}{\partial p_{dft}} \right)^{-1} s_{dft}.$$

We then replace $\partial s / \partial p$ with our estimate of its value to obtain $\hat{mc}$ and plug the result into the empirical analog of equation (5) to recover estimates of γ_d , γ_t , and F_ω .

We find that the average marginal cost of a drug in our sample is \$20.19. Heterogeneity exists across drugs and over time. For a sense of dispersion, we compute the expected log marginal cost of each drug, $\hat{\gamma}_d + \hat{\gamma}_t$, and plot the resulting densities for cartelized and uncartelized drugs separately. Supplemental Appendix Figure C.2 reports the result, which shows that the mean, variance, and “shape” of the two distributions are similar. To assess whether members and nonmembers have different costs, we plot the distribution of $\hat{\omega}$ separately for each group. Supplemental Appendix Figure C.4 reports the result, which shows that the two distributions closely resemble one another.

We use estimates of γ_d and γ_t to predict how marginal costs would have evolved in cartelized markets following cartel formation. In Supplemental Appendix C.4, we provide two pieces of evidence to support our extrapolation. One is goodness-of-fit for a leave-out sample. The other is that predicted profit margins match ones reported in audited financial statements (Teva Pharmaceuticals Industries 2011–2016). We also show that our estimates generate reasonable implied damages between 2013 and 2015.

²⁶Column 4—included mostly for completeness—shows that column 3’s results are equally as insensitive as column 1’s at distinguishing between authorized and ordinary generics. Since accounting for heterogeneity in price sensitivity is important, while separating out authorized generics is not, we rely on estimates from column 3 for the remainder of the paper.

C. Stage I: Sunk Costs

Method.—The necessary conditions of a simultaneous move Nash equilibrium imply bounds on sunk costs. If we observe cartel member f enter j , then

$$(12) \quad \sum_{d \in \mathcal{D}_j} VE_d^M(\chi^M, \chi^N) \geq \mathbb{E}[\theta_{jf} | \mathcal{I}_{jf}],$$

and if not, then

$$(13) \quad \sum_{d \in \mathcal{D}_j} VE_d^M(\chi^M + 1, \chi^N) < \mathbb{E}[\theta_{jf} | \mathcal{I}_{jf}].$$

If we observe nonmember f enter j , then

$$(14) \quad \sum_{d \in \mathcal{D}_j} VE_d^N(\chi^M, \chi^N) \geq \mathbb{E}[\theta_{jf} | \mathcal{I}_{jf}],$$

and if not, then

$$(15) \quad \sum_{d \in \mathcal{D}_j} VE_d^N(\chi^M, \chi^N + 1) < \mathbb{E}[\theta_{jf} | \mathcal{I}_{jf}].$$

Inequalities (12)–(15) are the basis for estimation.

We allow sunk costs to vary with two important observable factors. First, sunk costs depend on the number of strengths associated with the substance-delivery-release combination (e.g., Teva produces atorvastatin in four strengths: 10, 20, 40, and 80 milligrams).²⁷ When preparing an ANDA for filing, firms must demonstrate “bioequivalence” to the innovator drug at each strength level. The process is costly, as it involves measuring the time it takes for a given amount of the substance to reach the bloodstream in healthy volunteers. Separately, once approval is granted, most delivery methods require firms to design distinct packaging and install (or repurpose) separate equipment for each strength. Second, sunk costs vary with the delivery method. Compared to tablets and capsules, which are by far the most common dosage forms among orally administered medications, irregular delivery methods such as syrups, solutions, and chewables involve specialized equipment and more complicated packaging. Moreover, these methods are much more susceptible to bacteria growth, which necessitates sterile packing conditions. Finally, sunk cost depends on a disturbance that is unobserved by the econometrician but known by the firms when they decide whether to enter. We assume it has a symmetric distribution and is independent of the substance-delivery-release combination characteristics (i.e., the number of strengths and the delivery method).

²⁷ Production requires know-how that some firms may lack. To accurately analyze alternative policies that induce more entry, we require that a sufficient number of potential entrants possess this know-how. The requirement is easily met in our setting, which studies relatively “simple” drugs like orally administered solids (i.e., tablets and capsules), but it could bind in other settings.

Formally, we parameterize sunk costs such that

$$(16) \quad \theta_{jf} = \theta_0 + \theta_1 r_j + \theta_2 \ell_j + \eta_j.$$

Here, r denotes the number of additional strengths associated with the substance-delivery-release combination. ℓ is an indicator variable that equals one for drugs with an irregular delivery method. η_j represents the disturbance term known to the firms when they contemplate entry. This term permits the sunk costs of entering one market (e.g., warfarin tablets) to differ from the sunk costs of entering another (e.g., cefdinir capsules) in ways that influence behavior.

If the predictions of the model and the actions of the firms differ beyond the flexibility provided for in equation (16), then the differences are rationalized by mean-zero expectation errors, which we denote by ν_{jf} . Their presence reconciles a small number of cases where entry by a nonmember implies that a member should enter but does not (and vice versa). We assume that all potential member entrants realize the same expectational error and that the same is true for potential nonmember entrants. Given this assumption and the fact that these disturbances rationalize differences in the realized payoffs of cartel members relative to nonmembers, we can ascribe them to either group without loss of generality. We arbitrarily pin them to cartel nonmembers, defining $\theta_{jM} = \mathbb{E}[\theta_{jM} | \mathcal{I}_{jM}]$ and $\theta_{jN} = \mathbb{E}[\theta_{jN} | \mathcal{I}_{jN}] + \nu_j$.²⁸

Finally, we assume that $\hat{F}_D = F_D$ and that

$$(17) \quad \int_{\xi'} \int_{\omega'} \hat{\pi}_{d,t}^k(M, N, \xi, \omega) d\hat{F}_{\xi} d\hat{F}_{\omega} = \int_{\xi'} \int_{\omega'} \pi_{d,t}^k(M, N, \xi, \omega) dF_{\xi} dF_{\omega},$$

where $k \in \{(C, M), (C, N), B\}$. $\hat{F}_D$, $\hat{F}_{\xi}$, and $\hat{F}_{\omega}$ denote the empirical distributions of delays, demand shocks, and marginal cost shocks, and $\hat{\pi}_{d,t}^k(M, N, \xi, \omega)$ denotes $\pi_{d,t}^k(M, N, \xi, \omega)$ evaluated at our estimates of λ_d , λ_t , α_d , σ , γ_d , and γ_t rather than the true parameters. Assuming that expected profits are measured without error greatly reduces the computational burden required to bound sunk cost parameters and simulate counterfactual outcomes, making both exercises feasible for us.

Under these assumptions, we can construct moments based on inequalities (12) through (15) that produce consistent bounds on the parameters determining sunk costs (Ishii 2005; Ho 2009; Pakes et al. 2015; Wollmann 2018). The presence of η , which is observed by the firms but not the econometrician, complicates their construction. To economize on space, we present the moments below and leave our detailed discussion of how we solve the selection problem introduced by sunk cost disturbances to Supplemental Appendix E.

To estimate lower bounds, we compute

$$(18) \quad \frac{1}{J} \sum_j \frac{1}{\mu_j} \sum_{d \in \mathcal{D}_j} \frac{1}{2} \sum_{k \in \{M, N\}} \left[\widehat{VE}_d^k(\chi^M + \mathbf{1}_{\{k=M\}}, \chi^N + \mathbf{1}_{\{k=N\}}) \right. \\ \left. - \theta_0 - \theta_1 r_j - \theta_2 \ell_j \right] h_j^t < 0.$$

²⁸For readers more familiar with Pakes et al. (2015), our ν and η correspond to their $\nu_{1,-}$ and $\nu_{2,-}$ -type errors, respectively.

Moments indexed by ι are formed by interacting the bracketed term with a weight function, denoted by h_j^ι . The weight function includes a constant, an indicator for standard delivery method drugs, an indicator for nonstandard delivery method drugs, and indicators for substance-delivery-release combinations available in one, two, and more than two strengths. J denotes the number of unique substance-delivery-release combinations, and μ_j denotes the number of drugs associated with each substance-delivery-release combination. $\widehat{VE}_d^M(\cdot)$ denotes $VE_d^M(\cdot)$ evaluated at our estimates of $\pi(\cdot)$, F_ξ , F_ω , and F_D rather than the true values. $\widehat{VE}_d^N(\cdot)$ is defined analogously.

To estimate upper bounds, we require additional notation. Let L be the set of j with at least one entrant, J_L be the size of that set, and w^i be a positive valued function of r_j and ℓ_j . Also, define

$$(19) \quad VE_j^+ = \frac{1}{2} \sum_{d \in \mathcal{D}_j} \sum_{k \in \{M, N\}} \left[\widehat{VE}_d^k(\chi^M + \mathbf{1}_{\{k=M\}}, \chi^N + \mathbf{1}_{\{k=N\}}) \right],$$

which represents the average of the cartel members' and nonmembers' entry values in j . Finally, for each moment i , order j by their values of $w_j^i VE_j^+$, and let Ψ_{wVE} denote the set of j that corresponds to the $J - J_L$ smallest values. The second set of moments is then given by

$$(20) \quad \begin{aligned} & \frac{1}{J} \sum_{j \in L} \frac{1}{\mu_j} \sum_{d \in \mathcal{D}_j} \sum_{k \in \{M, N\}} w_j^i \left[\frac{\mathbf{1}_{\{\chi^k \geq 1\}} \widehat{VE}_d^k(\chi^M + \mathbf{1}_{\{k=M\}}, \chi^N + \mathbf{1}_{\{k=N\}})}{\mathbf{1}_{\{\chi^M \geq 1\}} + \mathbf{1}_{\{\chi^N \geq 1\}} + \mathbf{1}_{\{\chi^M=0, \chi^N=0\}}} \right. \\ & \quad \left. - \theta_0 - \theta_1 r_j - \theta_2 \ell_j \right] \\ & - \frac{1}{J} \sum_{j \in \Psi_{wVE}} \frac{1}{\mu_j} \sum_{d \in \mathcal{D}_j} \sum_{k \in \{M, N\}} w_j^i \left[\frac{\widehat{VE}_d^k(\chi^M + \mathbf{1}_{\{k=M\}}, \chi^N + \mathbf{1}_{\{k=N\}})}{2} \right. \\ & \quad \left. - \theta_0 - \theta_1 r_j - \theta_2 \ell_j \right] < 0. \end{aligned}$$

Our moments provide one-sided restrictions on the parameters of interest, the most informative of which are the greatest lower bound and the least upper bound. Since our inference procedure is based on maxima and minima, respectively, rather than averages, we cannot rely on the central limit theorem. To obtain 95 percent confidence intervals around the true parameters, we follow Andrews and Soares (2010).²⁹ We search over a three-dimensional grid of candidate vectors for

²⁹Tests based on inequalities depend on the degree to which the moments are binding. One consequence is that including uninformative moments (i.e., ones that are satisfied for a very wide range of parameters) typically widens confidence intervals. The main innovation of Andrews and Soares (2010) is a procedure for, loosely speaking, deciding which moments are sufficiently uninformative to be discarded. The authors have found moment selection to be incredibly important in other settings. We de-emphasize it here only because it does not have a big effect on our results.

TABLE 3—SUNK COST ESTIMATES

Variable	Parameter	95% confid. interval
Constant	θ_0	[0.25, 2.4]
Number of strengths	θ_1	[0.975, 2.175]
Indicator for irregular delivery	θ_2	[0.4, 6.6]
Observations		220
Moments		10
Minimum sunk cost of entry		1.325
Average sunk cost of entry		3.781
Maximum sunk cost of entry		13.925

Notes: Sunk cost estimates are reported in millions of dollars. The bounds reported in the rightmost column are intervals in which the true parameters lie 95 percent of the time. To calculate the minimum, mean, and maximum sunk costs across the substance-delivery-release combinations, we set θ_0 , θ_1 , and θ_2 equal to their respective midpoints.

$[\theta_0, \theta_1, \theta_2]$, and we invert an Anderson-Rubin type test at each one (Chernozhukov, Hong, and Tamer 2007).

Parameter Estimates.—Table 3 reports sunk cost estimates. The final column provides 95 percent confidence intervals, all of which exclude 0. The coefficient on the constant term, θ_0 , is bounded between \$250,000 and \$2.4 million. Reflecting the fact that sunk costs scale with the number of strengths per substance-delivery-release combinations, we find that θ_1 is bounded between \$975,000 million and \$2.175 million. Consistent with irregular delivery methods involving substantially higher sunk costs, we find that θ_2 is between \$0.4 million and \$6.6 million. Calculated at the midpoint of each identified set, entry costs range between \$1.325 million and \$13.925 million, depending on the characteristics of the substance-delivery-release combination. The average across drugs, weighting each equally, is \$3.78 million.

Our entry cost estimates align with statements made by agency officials and medical researchers. For example, in 2014, former FDA Commissioner Gottlieb stated, “Filing a generic application requires an average of about \$5 million and can cost as much as \$15 million” (Gottlieb 2016). His estimate is very close to the \$3.8 million mean and \$13.9 million maximum values we report in Table 3. Similarly, our estimates fall squarely within the bounds reported by Scott Morton (1999), who surveyed FDA officials around 1999 and found that sunk costs range from \$382,000 to \$31 million (in 2019 dollars, i.e., \$250,000 to \$20 million in the original text). Even more to the point, Baker-Smith et al. (2008) ran six bioequivalence studies between 1997 and 2004 and documented their per study expenditures. The authors spent between \$807,000 and \$1.25 million per study (in 2019 constant dollars), which aligns with our estimate of per strength entry costs.³⁰

³⁰ Putting aside factory setup, pharmacological studies are the main reason entry costs depend on the number of strengths, so they are an especially good benchmark for our estimates of the coefficient on r . Also note that while the range Baker-Smith et al. (2008) give is centered around the lower bound of our confidence interval, our figure also accounts for the equipment required to produce the drug.

VI. Testing Assumptions

Throughout the last two sections, we have assumed that once a cartel forms, members set collusive prices and nonmembers best respond in all subsequent periods.³¹ In this section, we test our assumptions. We first formally quantify the incentives of cartel members to collude, relying on definitions and methods proposed by Igami and Sugaya (2022). Following Backus, Conlon, and Sinkinson (2021), we then test whether the data are consistent with our conduct assumptions.

A. Incentives to Collude

To accurately characterize incentives to comply, we must pay special attention to timing. Throughout the estimation of the model and simulation of counterfactual outcomes, we assume each period lasts one year. This level of aggregation is common in the literature and reflects limitations imposed by data and computational costs. However, in our setting, cartel members can detect, and potentially retaliate against, noncompliance *very* quickly. For instance, in 2013, Glenmark undercut Teva in one of the cartelized markets, apparently due to miscommunication. The response was almost immediate. A confused Teva employee sent an email whose only contents were “???” to NP, and exactly five minutes later, NP replied, “I know ... made the call already.” The next day NP spoke to her counterpart at Glenmark, who rescinded the offer (*Connecticut et al. v. Teva Pharm. USA, Inc.* 2019, p. 134).³²

To reflect the speed at which punishment would occur without complicating notation throughout the rest of the paper, we maintain that all changes in market structure and environment occur annually, but we let cartel members punish noncompliance within two weeks.³³ We again note that detection is described as nearly instantaneous in the complaint, and hypothetical retaliation could quickly follow. Formally, we assume that first-stage decisions are made instantaneously and that the second stage of each period has S substages of equal duration, which are indexed by s . In this subsection only, we also assume that in every substage, firms set prices, while buyers draw new demand shocks, ϵ and ζ , and make purchase decisions. We emphasize that because all determinants of market structure, demand, and cost are fixed throughout the year, this assumption does not affect any other analysis in the paper.

We assume that prices charged in one substage are observed by the time the next one starts. The history of play at substage s of period t is then given by

$$(21) \quad h_d^{ts} = \left((p_{df\tau\psi}, p_{df\tau\psi}^{C,M})_{f \in \mathcal{M}_d, t_d \leq \tau \leq t-1, 1 \leq \psi \leq S}, (p_{df\tau\psi}, p_{df\tau\psi}^{C,M})_{f \in \mathcal{M}_d, 1 \leq \psi \leq s-1} \right).$$

³¹ Recall firms expect incumbents to respond “rationally” to changes in market structure. For instance, the Complaint describes firms “playing nice in the sandbox” as “responsible” or “rational” competitors (*Connecticut et al. v. Teva Pharm. USA, Inc.* 2019, p. 42). By contrast, the Complaint does not explicitly discuss punishments. We assume the simplest model of Nash reversion.

³² This event also speaks to how gravely members treated noncompliance. By withdrawing an attractive price, Glenmark upset the buyer, who was a very large customer.

³³ As we show below, ICCs are satisfied in our data. Nearly all exhibit significant slack. When the detection/punishment period is as long as two months, a small number of constraints are violated for extreme draws of ξ and ω .

In each cartelized market, all members must keep track of every price specified by the cartel agreement ($p_{dft\psi}^{CM}$) and charged by a member ($p_{dft\psi}$) since cartel formation. To ease exposition, we distinguish between two series of prices. The first corresponds to all substages in all periods between cartel formation and the last period, that is, between t_d and $t - 1$. The second corresponds to all substages between 1 and $s - 1$ in the current period, t . If in either series there is a point in time (τ, ψ) such that $p_{dft\psi} \neq p_{dft\psi}^{CM}$, then we call the cartel “noncompliant.” Otherwise, we call it “compliant.”

Consider any period $t \geq t_d$ and substage s in which the cartel is compliant. Suppose that the number of member and nonmember entrants that have arrived by this point are given by E_M and E_N , respectively, and that the firms have drawn demand and marginal cost shocks represented by vectors ξ and ω , respectively. The value to f of complying with the cartel agreement equals

$$\begin{aligned}
 (22) \quad V_{dfts}^C(E_M, E_N, \xi, \omega) &= \frac{1}{S} \sum_{\psi=0}^{S-s} \tilde{\delta}^\psi \pi_{d,f,t}^{CM}(M_{d,t} + E_M, N_{d,t} + E_N, \xi, \omega) \\
 &+ \tilde{\delta}^s \sum_{\tau=1}^{\infty} \delta^\tau \sum_{e_M=0}^{\chi_{Md}-E_M} \rho(\chi_{Md} - E_M, e_M, \tau) \sum_{e_N=0}^{\chi_{Nd}-E_N} \rho(\chi_{Nd} - E_N, e_N, \tau) \\
 &\times \int_{\xi', \omega'} \pi_{d,f,t+\tau}^{CM}(M_{d,t+\tau} + E_M + e_M, \\
 &N_{d,t+\tau} + E_N + e_N, \xi', \omega') dF_{\xi'} dF_{\omega'},
 \end{aligned}$$

where $\tilde{\delta} \equiv \delta^{1/S}$ and $\rho(\cdot)$ is defined in equation (8) above. The first term represents the discounted sum of cartel profits in the current period, while the second term represents the continuation value, which equals the discounted sum of expected cartel profits in all future periods. Expected profits are an average over four random variables: e_M , e_N , ξ' , and ω' . Since χ_{Md} denotes the total number of member entrants and E_M denotes the number that have already arrived, $\chi_{Md} - E_M$ denote the remainder (i.e., the number of firms that have not yet arrived). e_M denotes how many of the remaining member entrants have arrived by period $t + \tau$. e_N denotes the analogous number for nonmembers. ξ' and ω' denote demand and marginal cost shocks, respectively, that may be drawn in future periods.

The corresponding value of not complying equals

$$\begin{aligned}
 (23) \quad V_{dfts}^D(E_M, E_N, \xi, \omega) &= \frac{1}{S} \left[\pi_{d,f,t}^D(M_{d,t} + E_M, N_{d,t} + E_N, \xi, \omega) \right. \\
 &+ \left. \sum_{\psi=2}^{S-s} \tilde{\delta}^\psi \pi_{d,f,t}^B(M_{d,t} + E_M, N_{d,t} + E_N, \xi, \omega) \right] \\
 &+ \tilde{\delta}^s \sum_{\tau=1}^{\infty} \delta^\tau \sum_{e_M=0}^{\chi_{Md}-E_M} \rho(\chi_{Md} - E_M, e_M, \tau) \sum_{e_N=0}^{\chi_{Nd}-E_N} \rho(\chi_{Nd} - E_N, e_N, \tau)
 \end{aligned}$$

$$\times \int_{\xi', \omega'} \pi_{d,f,t+\tau}^B (M_{d,t+\tau} + E_M + e_M, \\ N_{d,t+\tau} + E_N + e_N, \xi', \omega') dF_{\xi'} dF_{\omega'},$$

where $\pi_{d,f,t}^D(M_{d,t} + E_M, N_{d,t} + E_N, \xi, \omega)$ denotes f 's "disagreement" payoff (i.e., the amount of money f earns in a period if all other members comply with the cartel agreement but f best responds). The first term represents the portion of the disagreement profit that the firm would receive, given that other firms retaliate in the following substage. The second term represents the discounted sum of Nash-Bertrand profits in the remainder of the current period, t , while the third term represents the sum of expected Nash-Bertrand profits in all future periods. The difference between $V_{dfts}^C(E_M, E_N, \xi, \omega)$ and $V_{dfts}^D(E_M, E_N, \xi, \omega)$ represents f 's incentive to comply with the cartel agreement.

A cartel is stable if all members' incentives are positive in the present moment and at all points in the future. Thus, to assess stability in drug market d at time (t, s) , one can check whether

$$(24) \quad \min_{\substack{(\tau, \psi): \\ (\psi \geq s, \tau = t) \vee \\ (\psi \leq s, \tau > t)}} \left\{ \min_{\substack{E_M: \\ E_M \leq \chi_{Md}}} \left\{ \min_{\substack{E_N: \\ E_N \leq \chi_{Nd}}} \left\{ \min_{\xi, \omega} \left\{ \min_{f \in \mathcal{M}_{d\tau}} \{V_{dft\psi}^C(E_M, E_N, \xi, \omega) \right. \right. \right. \right. \right. \\ \left. \left. \left. \left. \left. - V_{dft\psi}^D(E_M, E_N, \xi, \omega) \right\} \right\} \right\} \right\} \right\} \geq 0.$$

Inequality (24) depends on the minimum incentive to comply with the agreement obtained over the following: all remaining substages $\psi \geq s$ in the current period $\tau = t$; all subperiods ψ in all future periods $\tau > t$; all possible draws of delays, which determine the number of member and nonmember entrants that have entered by that point in time; all possible draws of demand and marginal cost shocks; and all firms in $\mathcal{M}_{d\tau}$, the set of active members in d at τ .

We follow Igami and Sugaya (2022) and evaluate cartel stability in the data using the following procedure. For each period τ , substage ψ , and cartelized drug d , we enumerate every possible combination of member and nonmember entrants that might have entered by that time, denoted by E_M and E_N , respectively. Then, for each combination, we draw a large number (500) of vectors of demand and marginal cost disturbances, denoted by ξ and ω , respectively. Next, we calculate the lower envelope of the difference between $V_{dft\psi}^C(E_M, E_N, \xi, \omega)$ and $V_{dft\psi}^D(E_M, E_N, \xi, \omega)$ over $f \in \mathcal{M}_{d\tau}$, ξ , ω , E_M , and E_N . If the value is positive for all $\tau > t_d$ and for all $\psi \geq s$ given $\tau = t_d$, then the cartel is stable; trigger strategies form an equilibrium. Finally, we plot the distribution of the lower envelope across drugs over time.

Figure 4 reports the result. Three features stand out. First, the least slack ICC, represented by the solid line, is positive at every point in time. In other words, all of the cartels we study are stable. Second, in anticipation of entry, many markets' ICCs exhibit steady declines in the first two years after cartel formation. Their slope and duration reflect how market structure relates to disagreement profit and how delays are distributed. The temptation to undercut a cartel derives from the comparatively

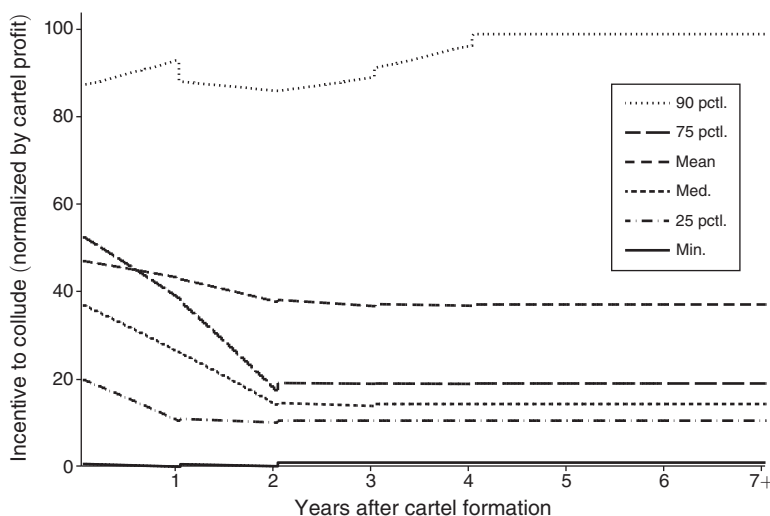


FIGURE 4. GENERIC DRUG CARTELS ARE STABLE

Notes: We compute the lower envelope of the difference between $V_{dfr,\psi}^C(E_M, E_N, \xi, \omega)$ and $V_{dfr,\psi}^D(E_M, E_N, \xi, \omega)$ over E_M, E_N, ξ, ω and cartel members, with $\tau = 0$ corresponding to the year the cartel is formed. To facilitate comparisons across drugs, we scale resulting values by a drug-specific factor equal to the mean cartel profit in the first substage of the first period following cartel formation (i.e., we divide through by the average $\pi_{d,fr,d}^C(E_M, E_N, \xi, \omega)/S$ over E_M, E_N, ξ, ω and cartel members). We then plot the distribution of resulting values across drugs along the y-axis and the number of years since cartel formation on the x-axis.

high markups that a member would earn on the business it would steal from others. Since members can typically steal more share in markets with more firms, entry increases this temptation. As a result, the weakest incentives to collude across all possible combinations of E_M and E_N correspond to all entrants arriving as soon as possible. Based on the empirical distribution of delays, this can occur as early as $\tau = 2$. Third, ICCs are otherwise fairly constant over time.

In practice, ICCs may have been satisfied by even greater margins than what we report here. As we stated above, we assume that each cartel agreement is unaffected by prices charged in other drug markets. However, statements published in the complaint can be interpreted as concerns about cross-market punishment, which typically increase slack in the most binding ICCs.³⁴

ICCs could also have been affected by detection risk. To be specific, firms could have feared that, with some positive probability, cartel discovery by antitrust authorities would cause collusion to break down, leading to an abrupt decline in prices and profits. Our analysis abstracts away from these concerns for two reasons. First, such fears would be unfounded in this setting. Cartel agreements were discovered and prosecuted, but collusion was sustained (see our conduct tests below). Second, if entrants harbored these fears, then the value of entry that we estimate would be

³⁴Least-slack constraints reflect asymmetries among members within markets. Cross-market pooling increases ICC slack. Noncompliance, that is, cheating, may also involve significant shame and/or guilt, which increases incentives to comply.

lower, as would the sunk costs we estimate. Yet our sunk cost estimates are in line with the expectations of industry participants (see the discussion in Section VIC). In many other markets, however, detection risk may be a first-order consideration, so identifying and measuring it remains an important area for empirical work.

B. Conduct Testing

We now test whether the conduct we observe in the data is consistent with the assumptions of our model. To do so, we rely on an intuitive test proposed by Backus, Conlon, and Sinkinson (2021), which adapts the nonnesting framework proposed by Rivers and Vuong (2002) to exclusion restrictions suggested by Berry and Haile (2014). The test provides pairwise comparisons but does not require that either behavioral model is correctly specified. It is based on the idea that if conduct is correctly specified, then variables that determine markups but do not influence marginal costs will be uncorrelated with marginal cost disturbances implied by that model.

Recall our maintained assumptions: All firms set competitive prices in uncartelized markets; all firms set competitive prices in cartelized markets before cartel formation; and after cartel formation, members set prices that maximize their joint profits, while nonmembers best respond. We conduct four tests. In each, we arbitrarily number our “baseline” model of conduct “1” and the comparison model “2.”

In Tests A and B, we assess whether, following cartel formation, members set prices that *maximize joint profit*. Model 2 is identical to Model 1 except that, after a cartel forms, members set prices that *maximize individual profits* in Model 2. To abstract away from the effect of the states’ investigation, Test A omits any observations after 2015. To study post-investigation behavior, Test B omits the period between 2013 and 2015. In Test C, we assess whether nonmembers comply with cartel agreements (i.e., they are “folded into” the cartel). Model 2 is identical to Model 1 except that, after a cartel forms, all firms set prices that maximize the joint profit of all firms (rather than only members setting prices that maximize their joint profit). In Test D, we assess whether member entrants comply with the cartel agreement. That is, Model 2 is identical to Model 1 except that only member incumbents set prices that maximize their joint profit (as opposed to all members setting prices that maximize their joint profit).

Each test involves the following steps. First, for Model 1, compute equilibrium markups, denoted by $\widehat{mk}_{dft}^1$, subtract them from observed prices to obtain $\widehat{mc}_{dft}^1$, and repeat this process for Model 2 to obtain $\widehat{mk}_{dft}^2$ and $\widehat{mc}_{dft}^2$. Second, for Model 1, estimate the empirical analog of equation (11) to obtain $\widehat{\omega}_{dft}^1$, and repeat this calculation for Model 2 to obtain $\widehat{\omega}_{dft}^2$. Third, let z_{dft} denote a vector that includes drug fixed effects, year fixed effects, and an indicator variable that takes a value of one if and only if the complaint alleges the price of d was cartelized in or before year t . Then, regress $\widehat{mk}_{dft}^1 - \widehat{mk}_{dft}^2$ on z_{dft} and obtain predicted values, denoted by $\widehat{g}(z_{dft})$. Fourth, let $\widehat{Q}_1$ denote the square of the average value of $\widehat{\omega}_{dft}^1 \widehat{g}(z_{dft})$, let $\widehat{Q}_2$ denote an analogous value for Model 2, and compute $\widehat{Q}_1$ and $\widehat{Q}_2$. Fifth, repeat steps 1–4 on a large number (500) of bootstrapped samples and compute the standard error of the difference between $\widehat{Q}_1$ and $\widehat{Q}_2$, denoted by $\widehat{\varsigma}$. Sixth, compute the test statistic, which

TABLE 4—RESULTS OF CONDUCT TESTS

	$\tilde{Q} \times 100$		Test	
	Baseline	Alternative	statistic	p-value
Test A. Model versus competition (pre-investigation)	0.27165	0.36047	−3.49	< 0.0001
Test B. Model versus competition (post-investigation)	0.41182	0.62954	−4.73	< 0.0001
Test C. Model versus nonmembers comply	1.1801	3.2849	−5.59	< 0.0001
Test D. Model versus member entrants do not comply	0.00036	0.00037	−4.1	< 0.0001

Notes: This table reports the results of the testing procedure proposed by Backus, Conlon, and Sinkinson (2021) for pairwise comparisons described in the text. The test statistic is distributed standard normal. The standard error of the difference between $\tilde{Q}_1$ and $\tilde{Q}_2$ is obtained via bootstrapping.

equals $\hat{\zeta}^{-1}(\tilde{Q}_1 - \tilde{Q}_2)$ and is distributed standard normal. The null hypothesis is that both models describe the data equally well. A negative test statistic indicates that the data favor our Model 1, while a positive one indicates that the data favor Model 2.

Table 4 reports the results. To interpret them, recall that the test statistic is distributed standard normal and that smaller values imply that the data fit Model 1 better than Model 2. Each test strongly favors Model 1: Test A supports the government's claim that defendants conspired to fix prices through at least 2015, while Test B implies collusion continued for the remainder of the sample, albeit perhaps in a tacit form. Test C indicates that nonmembers are not folded into the cartel. The finding is consistent with nonmembers undercutting members; we further note that NP formed cartels in markets where almost all other firms were members. (The test further demonstrates that potential entrants were attracted to supernormal *equilibrium* profits, not simply high prices; consistent with Caoui (2022), entrants price less aggressively when facing the cartel.) Test D implies that member entrants complied with cartel agreements, which is consistent with how carefully NP coordinated the entry of a member entrant into cartelized markets.

VII. Counterfactual Analysis

In this section, we measure the effect of entry on the cartelized markets. We also evaluate alternative FDA policies that attract additional entrants and/or permit them to enter sooner.

A. Effects of Equilibrium Entry

To assess how important entrants are in cartelized drug markets, we compare the predictions of a model that allows for entry following cartel formation with the predictions of a model that prohibits it. To arrive at the first set of predictions, we let the market structure evolve according to observed entry. For each cartelized market d and year t following cartel formation, we draw a large number (1,000) of vectors of demand and marginal cost shocks, and for each draw, we recompute the equilibrium of the second stage. For the second set of predictions, we replicate this process but remove all products associated with ANDAs filed after cartel formation. This eliminates 207 entrants spread over 69 markets. To summarize our findings, we compute the sum of profits by year and plot the difference between the first and second model.

Panel A of Figure 5 reports the result in event time, with year zero corresponding to cartel formation. Entrants arrive in markets as early as three years after cartel formation and almost immediately exert downward pressure on profit, reducing it by an average of \$60 million, or about 10 percent. More entrants arrive over the following four years, so that within seven years of cartel formation, entry reduces profit by an average of \$140 million, or 23 percent. Measured at year 0 and under the assumption that cartel members continue to comply with the agreement, albeit perhaps tacitly, entry reduces the present value of firms' profits by just under \$1 billion. Moreover, consumer surplus rises more than second-stage profit falls, as many buyers switch away from the outside option as prices fall.

The counterfactual exercise also speaks to potential deterrence. Deciding whether to fix prices requires weighing supranormal profits against coordination costs and legal risks. Entry reduces the supranormal profits without reducing the risks and coordination costs, so it may deter cartel formation altogether in some settings, even though it did not do so here. To understand the effect it can have on incentives to reach a collusive agreement, we replicate panel A of Figure 5 but plot the profit earned by *incumbent cartel members* (rather than by *all firms*).

Panel B of Figure 5 reports this result (note that the scale of the axes has changed). Since the profit measured in panel A includes profit earned by entrants, which is irrelevant to incumbent cartel members, the profit measured in panel B declines even more steeply as entrants arrive. Within seven years of cartel formation, entry reduces the amount cartel members earn by nearly half. In light of this sharp drop, it is easy to see how entry might deter some cartels from forming in the first place, even if it was insufficient deterrence in this instance.

B. Effects of Alternative Policies

Our estimates imply that even when entrants face large sunk costs and long regulatory delays, they can still significantly affect equilibrium outcomes. This finding naturally raises the question of whether society would benefit from policies that encourage more entrants and hasten their arrival. We examine two sets of policy changes. First, we reduce the cost of entry by between \$150,000 and \$600,000, a 4–16 percent decline for the average substance-delivery-release combination. Second, we reduce regulatory delays by between 0.5 and 2 years.

The FDA could implement these changes.³⁵ One option is for the agency to dramatically reduce sunk costs by cutting fees, which are substantial. At the time of writing, manufacturers must pay a base rate of \$1.5 million per year, \$200,000 per factory per year, \$225,712 to file an ANDA, and another \$75,000 per drug to ensure the details of production remain confidential. Other options may reduce costs associated with entry without reducing the revenue received by the agency. Recent proposals include publishing “best practices for manufacturers in submitting an ANDA” and “enhanc[ing] communication between manufacturers and FDA”

³⁵ Some of these changes may require the agency to reduce or redirect resources. No part of this discussion should be taken to mean that the time and expense associated with generic drug evaluations or any other FDA oversight activities are wasteful. That is, our analysis implies that lower fees and faster approvals generate significant surplus for buyers when incumbents are cooperating, but these benefits must be compared to costs, which are beyond the scope of the present paper to forecast accurately.

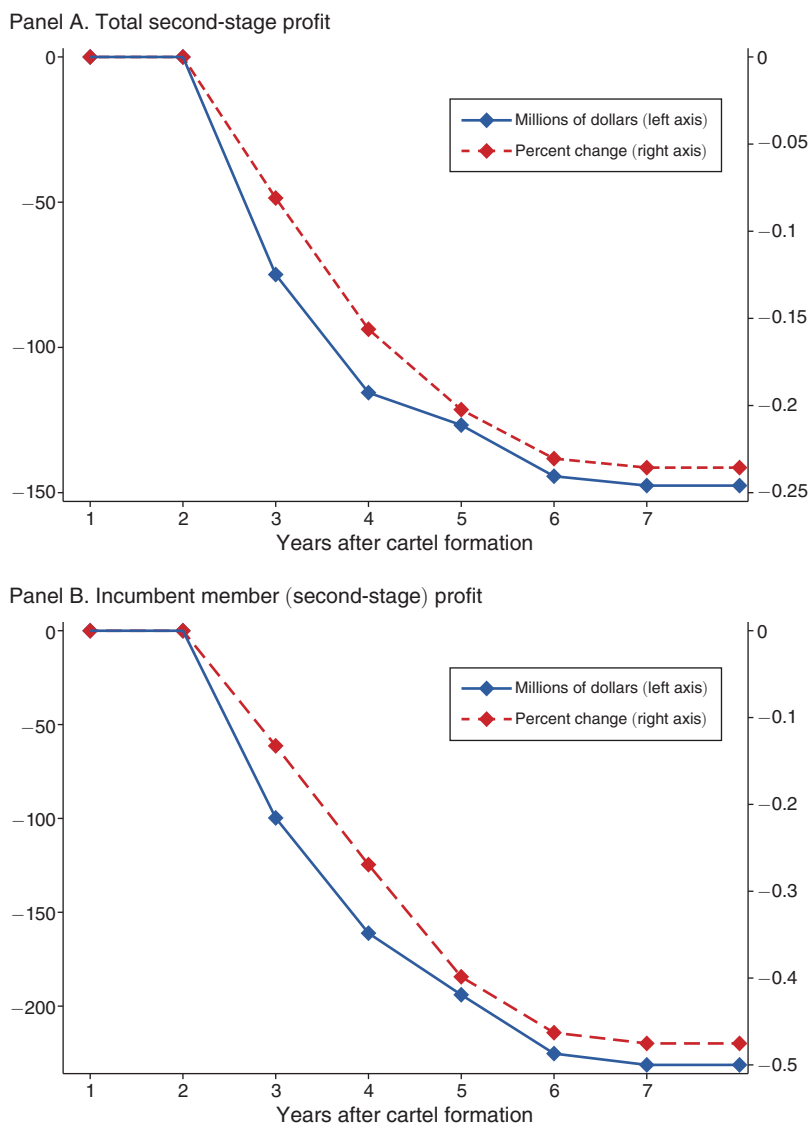


FIGURE 5. EFFECTS OF EQUILIBRIUM ENTRY ON FIRMS' PROFIT

Notes: This figure reports the difference in profit due to observed entry on the y-axes against the number of years since cartel formation on the x-axes. Primary y-axes measure dollar-denominated differences, while secondary y-axes measure percentage point differences (relative to profit earned under the no-entry model). Panel A reflects profits earned by all firms, whereas panel B reflects profits earned by incumbent cartel members.

(Waxman et al. 2018). To shorten delays, the agency could reallocate staff to prioritize timely reviews. Notably, the agency has experimented with expedited evaluations, proving that fast approval is feasible (presumably without compromising the integrity of the process). For instance, the “Prioritization of the Review of Original ANDAs, Amendments, and Supplement” policy, introduced in 2020, targets four- to eight-month approval for certain filings.

TABLE 5—EFFECTS OF REDUCING ENTRY COSTS AND DELAYS

	Sunk costs reduced by				Delays shortened by			
	\$0.15	\$0.3	\$0.45	\$0.6	0.5	1	1.5	2
Mean sunk cost	4.3	4.15	4.0	3.85	4.45	4.45	4.45	4.45
Mean delay	5.1	5.1	5.1	5.1	4.6	4.1	3.6	3.1
Drugs with additional entry	14	27	35	46	15	24	41	41
Additional entrants	17	38	56	85	18	28	53	63
Change in agency revenue (I)	−37.8	−75.6	−113.4	−151.2	—	—	—	—
Change in agency revenue (II)	22.27	58.88	90.95	146.28	21.05	35.22	68.3	78.88
Changes in total profits	−64.46	−153.05	−211.86	−301.56	−209.66	−373.57	−665.21	−826.34
Consumer comp. variation	33.27	62.23	90.06	156.28	308.17	611.72	1,119.26	1,470.0

Notes: Columns correspond to distinct counterfactual policy experiments. Columns 1–4 report the result of reducing the sunk entry costs, while columns 5–8 report the result of shortening entry delays. “Change in agency revenue (I)” represents lost revenue due to fee reductions, which we assume account for one-half of entry cost reductions. “Change in agency revenue (II)” represents revenue gained from additional filing fees. We assume that the FDA collects one-quarter of entry costs, less fee reductions. Dollar-denominated figures are reported in millions, while delays are reported in years.

These policy alternatives map to changes in θ_0 or F_D . For each of them, we adjust the primitives, simulate market outcomes, and report how these differ from the status quo. We recompute equilibrium entry in the first stage and equilibrium prices in the second stage, integrating out over the distributions of delays and of demand and marginal cost disturbances. We also recompute the ICCs to ensure they are satisfied (they are in every instance). Multiple equilibria may arise in the first stage, so our simulations must select among them. To obtain a single Nash equilibrium in markets where multiple equilibria exist, we choose the one whose ratio of member to nonmember entrants is as close as possible to the average ratio we observe in the data following cartel formation. (Supplemental Appendix E provides more details.)

Table 5 reports the result. Reducing the sunk costs of entry attracts as many as 85 new entrants. Compensating variation rises by at most \$156 million, and profits decline by at most \$302 million. Assuming fees account for one-half of entry cost reductions, FDA revenue falls by around \$22–\$146 million. For instance, with a \$600,000 cost reduction, additional entry raises FDA revenue by \$146 million, but fee cuts reduce FDA revenue by \$151 million.

Bringing products to market faster has much greater overall effects. Even a half-year reduction in delays produces roughly the same changes in profits and consumer compensating variation as a \$600,000 reduction in sunk costs. From the perspective of patients and payers, two mechanisms merit consideration. First, when nonmembers arrive, they undercut the cartel in an effort to steal share, and the cartel responds by also lowering prices, so short delays mean fewer periods with monopoly prices. Second, shorter delays allow entrants to recoup their sunk investments in entry faster, which encourages additional entry.

Equally striking, the magnitudes of the changes scale roughly linearly with the duration of the reduction. Bringing products to market two years faster cuts industry

profit by over \$826 million and equates to \$1.5 billion in consumer compensating variation. The extent of additional entry is on par with a \$450,000 cost reduction, highlighting the relative importance of the first mechanism: It is essential that monopoly-like prices charged by the cartel are disciplined as quickly as possible.

VIII. Conclusion

We investigated the likelihood and impact of entry in the context of the largest price-fixing case in US history, which involved US generic drugs. Our data suggest that cartel formation successfully raised prices sharply but also attracted significant entry, even though entrants faced regulatory delays of several years. The facts of the case and patterns in the data indicate that cartel members were folded into existing cartels, while nonmembers were not.

To accurately forecast how market participants would behave under alternative behavioral assumptions and regulatory policies, we estimated a structural model that allows firms to endogenously enter and set prices, and we used the resulting estimates to simulate equilibrium outcomes. We found that cartel members have strong incentives to comply with the collusive agreements, which may partly explain why we also found that price-fixing persisted long after the states launched their investigations. Our estimates confirm that entry is not just slow but also expensive, with sunk entry costs ranging from \$1.3 million to \$13.9 million per substance-delivery-release combination.

We simulated counterfactual outcomes absent entry and found that cartel profits would have been 27 percent higher over our sample period absent entry. This means that entry protected consumers from facing even higher prices, even if it was incomplete and delayed. Since policymakers often have considerable discretion over costs and delays, we also simulated counterfactual outcomes under alternative regulatory environments. Our results indicate that reducing either entry costs or delays can improve consumer surplus, although the latter is especially effective.

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